

Multifunctionality in Janus CrMCN₄ (M = Si/Ge) Monolayers: Valleytronic Physics, Piezoelectric Response, and Photocatalytic Potential

Jubair Hossan Abir¹, M.R.U.S. Shapon¹, S.S.B. Pallab², Tanvir Khan¹,
Raihana Shams Islam^{1*}, Saleh Hasan Naqib^{1*}

¹Department of Physics, University of Rajshahi, Rajshahi 6205

²Department of Electrical and Electronic Engineering, University of Rajshahi, Rajshahi 6205

*Corresponding authors; Emails: jsr_eda1@ru.ac.bd (Saleh Hasan Naqib), rsislam@ru.ac.bd (Raihana Shams Islam)

Abstract

Two-dimensional Janus semiconductors integrating spin-valley coupling, piezoelectricity, and tunable optical responses offer a platform for multifunctional nanodevices. Here, first-principles calculations reveal complementary composition and strain effects in CrSiCN₄ and CrGeCN₄ monolayers. Both are found to be nonmagnetic direct-gap semiconductors, with gaps of 1.23 and 1.09 eV using the Perdew-Burke-Ernzerhof functional including spin-orbit coupling (PBE+SOC), respectively. HSE06 hybrid-functional calculations retain the direct-gap character, yielding gaps of 1.46 eV for CrSiCN₄ and 1.19 eV for CrGeCN₄. Cr-*d*/N-*p* hybridization governs the band edges, with valence-band spin splitting of 53.3 and 52.4 meV and conduction-band splitting about 1.6 meV. Opposite out-of-plane spin character and Berry curvature emerge at K and K'. Despite similar valley-edge spin splitting, CrGeCN₄ exhibits Berry-curvature hot spots of order 10³ bohr², suggesting enhanced multiband contributions. Biaxial strain from -4% to +4% reduces the respective gaps from 1.59 to 0.86 eV and 1.44 to 0.63 eV, drives direct-to-indirect transitions, and redshifts absorption while preserving valley spin contrast. CrGeCN₄ exhibits larger in-plane piezoelectric response ($d_{11} = 5.084$ pm V⁻¹), whereas CrSiCN₄ exhibits larger out-of-plane magnitude ($|d_{31}| = 0.082$ pm V⁻¹). Band-edge alignment satisfies the oxygen evolution reaction (OER) requirement but provides insufficient driving force for the hydrogen evolution reaction (HER), motivating further strain-induced band-edge modulation toward overall water splitting. These results establish composition and strain as complementary controls for tailoring Cr-based Janus monolayers for valleytronic, optoelectronic, electromechanical, and photocatalytic applications.

Keywords: Two-dimensional Janus semiconductors; Valleytronics; Berry curvature; Piezoelectricity; Strain engineering; Photocatalytic water splitting

1 Introduction

In two-dimensional (2D) semiconductors, the valley degree of freedom complements charge and spin as an additional carrier label that can be used for information processing [1]. Their strong in-plane bonding and small thickness also permit substantial elastic deformation, making strain a practical route for modifying electronic and optical properties [2,3]. For these applications, the orbital, spin, and momentum character of the band-edge states can be as important as the magnitude of the band gap.

Valleytronics uses inequivalent extrema in momentum space as an information degree of freedom. In hexagonal monolayers, K and K' form a time-reversal-related valley pair. When inversion symmetry is broken, the two valleys can carry opposite Berry curvature and orbital magnetic moment; spin-orbit coupling (SOC) can couple spin and valley and generate valley-contrasting spin splitting and polarization dependent optical selection rules [4]. In monolayer MoS₂, valley-selective circular dichroism follows from symmetry-controlled optical selection rules [5], optical helicity can create a valley population imbalance [6], and intrinsic valley Hall transport is observed in non-centrosymmetric odd-layer MoS₂ but suppressed when inversion symmetry is restored [7]. A microscopic description of a candidate valley material therefore requires its symmetry, orbital character, SOC response, and Berry curvature to be considered together.

Mechanical strain perturbs these quantities by changing lattice geometry, orbital overlap, and the relative energies of competing band extrema. In 2D semiconductors, tensile or compressive strain can shift band edges, change direct/indirect gap ordering, alter carrier transport, and move optical transition energies [2,3]. Relativistic calculations for monolayer transition-metal dichalcogenides further show that biaxial strain can reorganize orbital contributions and SOC-induced spin splittings [8]. Recent studies also demonstrate strain control of valley-related responses in magnetic or alloyed systems: V-alloyed MoTe₂ shows strain and electric-field dependent valley polarization [9], ferrovalley NbX₂/TaX₂ monolayers exhibit strain-tunable valley polarization together with Berry-curvature derived anomalous Hall response [10], and VSe₂/VN shows strain-dependent spin filtering and valley splitting alongside an anomalous Hall response [11]. These examples motivate treating strain as a controlled perturbation of the orbital-SOC valley structure rather than only as a band-gap tuning parameter.

The chemical-vapor-deposition synthesis of septuple-layer MoSi₂N₄ and WSi₂N₄ established a two-dimensional structural motif that does not depend on exfoliation from a layered bulk parent [12]. The architecture can be described as an MZ₂-like core embedded between A-Z sublayers and has been generalized to a broad MA₂Z₄ family with diverse compositions and electronic phases [13]. Subsequent surveys document electronic, magnetic, thermal-transport, and superconducting behavior across this family, with strain and elemental substitution providing substantial tunability [14]. Chemical substitution and Janus construction can lower the symmetry and create chemically inequivalent out-of-plane environments within the same septuple-layer framework [15]. This combination of compositional flexibility and symmetry control makes MA₂Z₄ derivatives well suited for examining how orbital character and electrostatic asymmetry influence SOC and valley physics.

Valley-related phenomena are already established in several MA₂Z₄ systems. MoSi₂N₄, WSi₂N₄, and MoSi₂As₄ exhibit spin-valley coupling, valley-contrasting Berry curvature, and circularly selective optical response [16]. Calculations for MoSi₂X₄ (X = N, P, As) show that strain can tune the band structure and SOC-driven valley features [17], while studies of WSi₂N₄-derived monolayers connect structural symmetry and orbital character with distinct spin-orbit fields and valley responses [18]. Janus derivatives add an out-of-plane asymmetric environment: MSiGeN₄ (M = Mo, W) has been predicted to show Rashba-type SOC splitting [19], and Janus WSiGeZ₄ systems display symmetry-dependent Berry curvature and spin texture [20]. Magnetic MA₂Z₄ members provide complementary examples, including spin-valley-coupled VSi₂N₄ [21] and

CrSi₂N₄/CrSi₂P₄, where SOC-driven valley spin splitting can be tuned by strain [22]. Taken together, these studies show that the valley response is controlled by the symmetry and orbital composition of the band edges, not by Janus asymmetry alone.

Cr-based septuple layers provide a particularly relevant reference point. First-principles calculations on CrC₂N₄ established a dynamically and thermally stable direct-gap semiconductor with high stiffness, carrier mobility, visible-light absorption, and piezoelectric response [23]. A later SOC-resolved study showed that the K/K' band edges possess opposite out-of-plane spin character, with frontier states dominated by Cr-*d* and N-*p* hybridization, and reported strain-dependent changes in Berry curvature and optical response [24]. These results identify CrC₂N₄ as a useful parent system for examining how chemical differentiation of one outer sublayer modifies Cr-centered SOC and valley physics.

Janus CrSiCN₄ and CrGeCN₄ were subsequently proposed by replacing the C atoms on one side of CrC₂N₄ with Si or Ge. Their structural stability, direct-gap electronic structure, Cr/N-dominated frontier states, band-edge charge distributions, electrostatic-potential differences, work functions, carrier effective masses and mobilities, strain-dependent band edges, optical absorption, and photocatalytic band alignment have already been characterized [25]. The same study reported spin-degenerate band structures and PDOS and therefore omitted spin coupling from the subsequent calculations [25]. Consequently, the SOC-resolved band-edge electronic structure, the magnitude and orbital origin of spin splitting at the K/K' valleys, the valley-resolved spin character, and the Berry-curvature response of these Janus monolayers were not addressed in that study. The evolution of these quantities and their response to Si-to-Ge substitution therefore remain to be clarified.

To address these research gaps, first-principles calculations were performed on pristine and biaxially strained Janus CrSiCN₄ and CrGeCN₄ monolayers. Their structural, electronic, piezoelectric, valleytronic, and optical properties were systematically investigated, including orbital contributions, Bader charge transfer, work function, Berry curvature, SOC-induced spin splitting, and carrier effective masses. Particular emphasis was placed on the interplay between frontier-state orbital character, valley-dependent Berry curvature, and strain-tunable electronic transitions.

2 Computational Methodology

First-principles calculations were performed using density functional theory (DFT) as implemented in the Vienna Ab-initio Simulation Package (VASP) [26,27]. The electron-ion interactions were described through the projector augmented-wave (PAW) method [27]. Exchange-correlation effects were treated using the Perdew-Burke-Ernzerhof (PBE) functional [28] within the generalized gradient approximation (GGA). The Heyd-Scuseria-Ernzerhof hybrid functional (HSE06) [29,30] was additionally employed to obtain more accurate electronic band gaps.

The monolayer was modeled under periodic boundary conditions with a vacuum region of approximately 25 Å along the surface normal. This separation minimized artificial coupling between periodic images. The Numerical convergence criteria was established by using a tested plane-wave kinetic-energy cutoff of 500 eV and the Brillouin zone sampling resolution of $2\pi \times$

0.03 Å⁻¹. During relaxation, the atomic positions and in-plane lattice parameters were optimized, while the out-of-plane lattice parameter was fixed. A self-consistent dipole correction was applied at the center of the vacuum region to account for the intrinsic polarization and its associated electrostatic asymmetry. The van der Waals (vdW) interactions between layers were incorporated through the DFT-D3 dispersion correction to the total energy [31]. The convergence tolerance is 10⁻⁵ eV for total energy and 0.02 eV/Å for all forces. Charge redistribution was quantified using Bader charge analysis [32].

The response to biaxial strain was investigated by uniformly varying the lattice parameters in the xy-plane. The applied strain was defined as,

$$\varepsilon = \frac{a - a_0}{a_0} \times 100\%, \quad (1)$$

where a_0 and a are the equilibrium and strained lattice constants, respectively. For each applied strain the internal atomic coordinates were reoptimized while the lattice parameters were held at their prescribed values. Fully relativistic noncollinear calculations incorporating spin-orbit coupling were subsequently performed on the relaxed structures. Brillouin zone sampling employed a (24×24×1) k-point mesh for both the self-consistent calculations and the non-self-consistent calculations used for Wannier interpolation. Maximally localized Wannier functions were generated with Wannier90 [33] and used to evaluate Berry curvature and associated quantities in WannierBerri [34]. The elastic stiffness tensor and piezoelectric stress tensor are directly calculated by the DFPT method in the VASP package. The optical properties were derived from the frequency dependent complex dielectric function to examine the evolution of the optical response under strain. Since electron-hole interactions were not explicitly treated through the Bethe-Salpeter equation, the calculated spectra were used primarily to assess the evolution of the optical response under strain.

3 Results and Discussion

3.1 Structural Properties of Monolayer CrMCN₄ (M = Si/Ge)

The optimized geometries of monolayer CrCSiN₄ and CrCGeN₄ exhibit hexagonal in-plane lattices with seven atoms per primitive cell, arranged into seven distinct atomic layers along the out-of-plane direction (shown in **Figure 1**). Their stacking sequences are N-Si-N-Cr-N-C-N and N-Ge-N-Cr-N-C-N, respectively. In both structures, the central Cr layer is sandwiched between chemically inequivalent N-C-N and N-Si-N or N-Ge-N sublayers, giving rise to an asymmetric septuple-layer architecture. The optimized in-plane lattice constants are $a = b = 2.689$ Å for CrCSiN₄ and 2.743 Å for CrCGeN₄, with an angle of approximately 120° between the in-plane lattice vectors. Replacing Si with Ge therefore increases the equilibrium in-plane lattice constant by approximately 2.03%, while preserving the seven-layer stacking arrangement. This value is in excellent agreement with the previously reported value [25]. The increment in the lattice parameter results mainly from the larger ionic radius of Ge compared to Si. Previous investigations have

confirmed the mechanical, dynamical, and thermodynamic stability of this structure. The calculated equilibrium structural parameters are summarized in [Table 1](#).

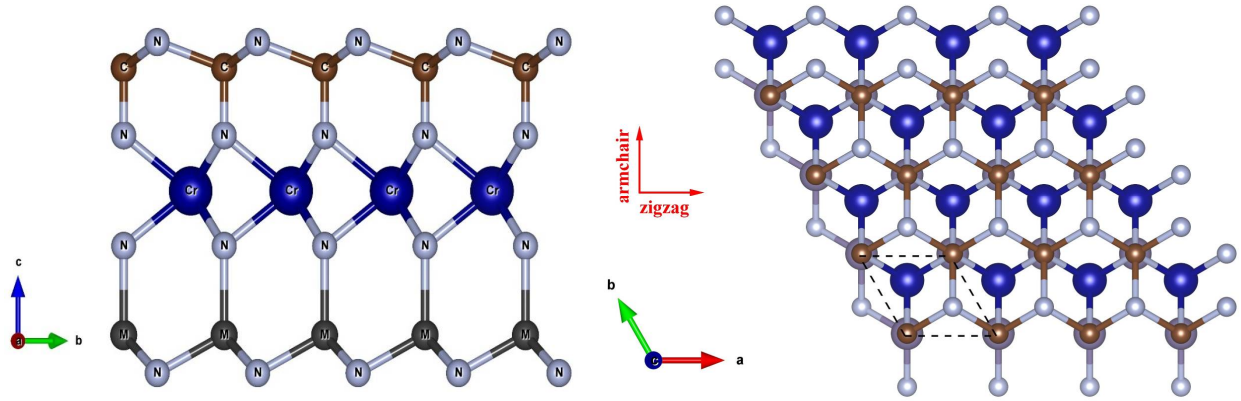


Figure 1. Optimized geometrical structures of monolayer CrMCN_4 ($M = \text{Si}$ and Ge). The side view (left) and top view (right) depict the atomic configuration of the monolayers, showing the arrangement of Cr, C, N, and Si/Ge atoms. The corresponding unit cell and crystallographic directions are also indicated.

Table 1. Structural parameters of monolayer CrSiCN_4 , CrGeCN_4 and CrC_2N_4 . Here, a is the in-plane lattice constant, $d_{\text{Cr-N(in)}}$ and $d_{\text{Cr-N(out)}}$ represent the two inequivalent Cr-N bond lengths, $d_{\text{C-N(out)}}$ and $d_{\text{Si/Ge-N(out)}}$ denote the corresponding C-N and Si/Ge-N bond lengths, respectively, and t_{eff} is the effective thickness of the monolayers.

Compound	a (Å)	$d_{\text{Cr-N(in)}}$ (Å)	$d_{\text{Cr-N(out)}}$ (Å)	$d_{\text{C-N(out)}}$ (Å)	$d_{\text{Si/Ge-N(out)}}$ (Å)	t_{eff} (Å)	Reference
CrSiCN_4	2.69	1.94	1.94	1.42	1.74	6.65	This work
CrGeCN_4	2.74	1.96	1.96	1.41	1.86	6.92	This work
CrC_2N_4	2.51	1.89	1.44	1.54	-	9.40	[24]
CrSiCN_4	2.70	-	-	-	-	-	[25]
CrGeCN_4	2.76	-	-	-	-	-	[25]

3.2 Charge Density and Bader Charge Analysis

The band-decomposed charge densities of the valence band maximum (VBM) and conduction band minimum (CBM) for CrSiCN_4 and CrGeCN_4 are presented in [Figure 2](#). The band-edge charge distributions are primarily localized around the Cr atoms and their neighboring N atoms, whereas the contribution from C atoms is comparatively negligible. This observation is consistent with the orbital-resolved electronic structures, confirming that the major bonds are mainly governed by Cr-d and N-p orbital interactions, as discussed in detail in the following section.

To further elucidate the charge redistribution and bonding characteristics, Bader charge analysis was performed using the converged charge density. The results reveal substantial charge transfer from Si/Ge, Cr, and C atoms toward the more electronegative N atoms. For CrSiCN₄, the Si, Cr, and C atoms lose 2.217e, 1.374e, and 0.697e, respectively, whereas the outer and inner N atoms accumulate 2.012e and 2.378e, respectively. The larger charge accumulation on the inner N atoms indicates their stronger involvement in the Cr-N bonding network. A similar charge-transfer behavior is observed in CrGeCN₄. These results demonstrate that both monolayers possess pronounced charge polarization with mixed ionic-covalent bonding characteristics, primarily originating from electron redistribution between electropositive Si/Ge, Cr, C atoms and electronegative N atoms.

3.3 Electronic Properties of the CrMCN₄ (M = Si/Ge) Monolayer

The electronic structures of monolayer CrMCN₄ (M = Si/Ge) were investigated using HSE06 band structures (without SOC), projected density of states (PDOS), PBE+SOC band structure with spin expectation value along the z direction and orbital-resolved band dispersions, as presented in [Figure 2-4](#).

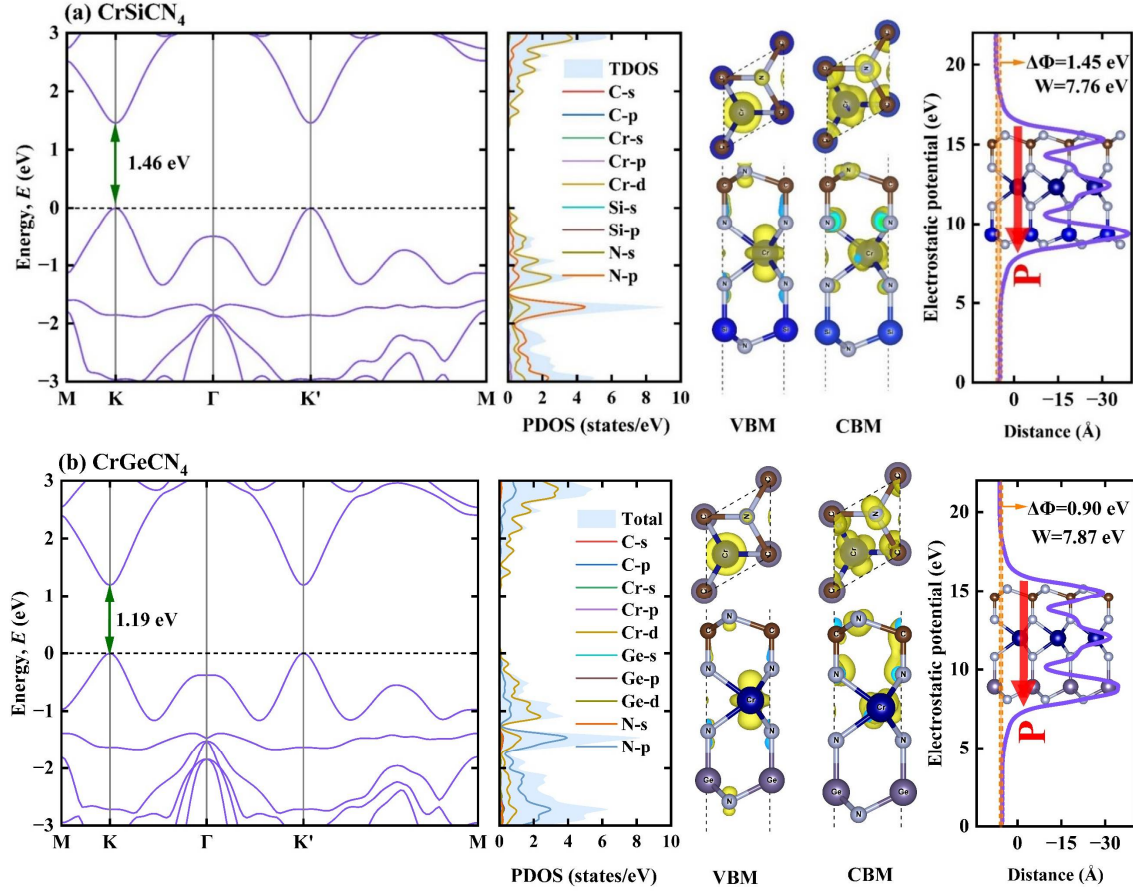


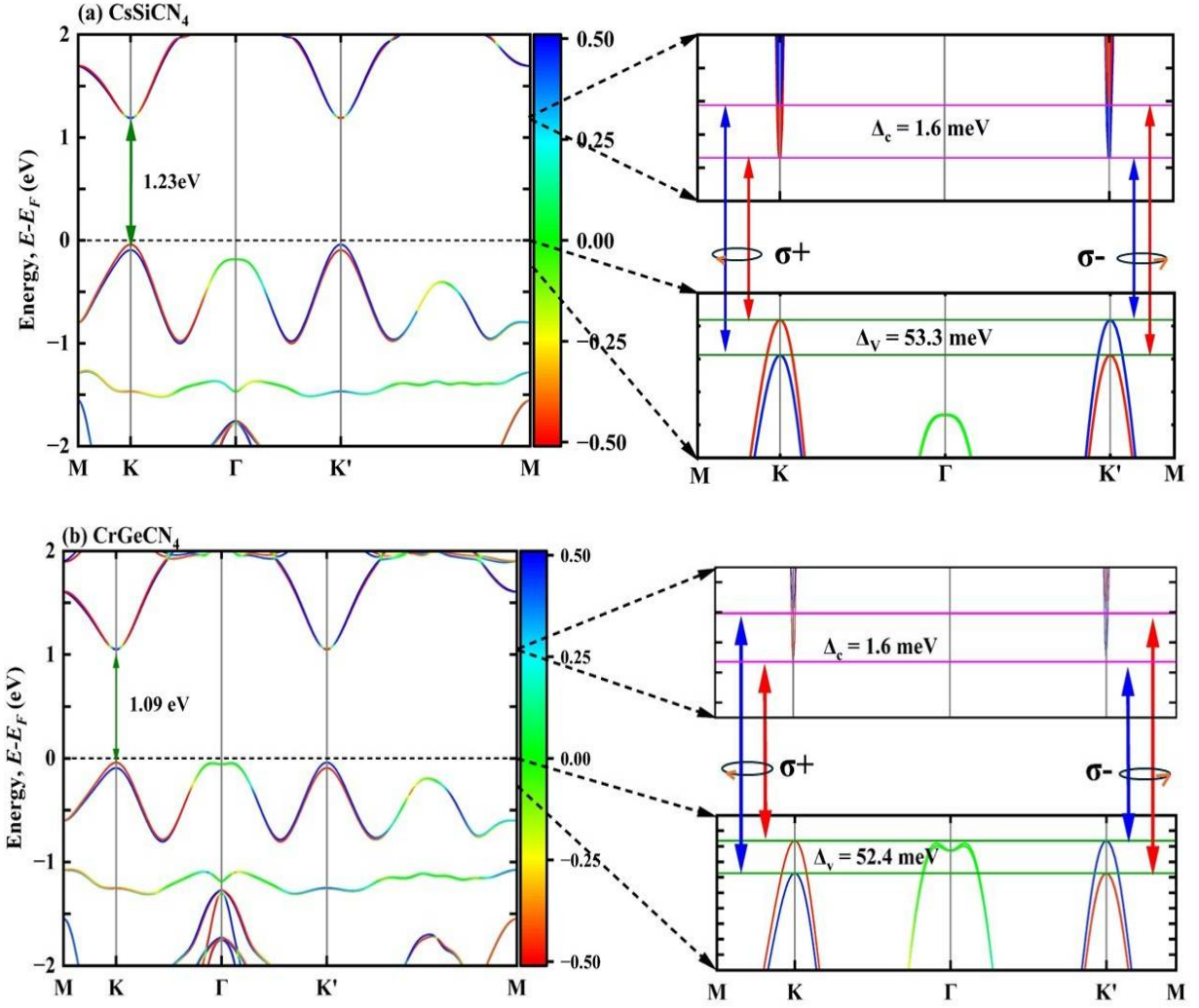
Figure 2. Electronic properties of monolayer CrSiCN₄ and CrGeCN₄ calculated using the HSE06 functional without spin-orbit coupling (SOC). (a, b) Electronic band structures of CrSiCN₄ and CrGeCN₄, respectively. The corresponding projected density of states (PDOS) is shown alongside the band structures. The band-decomposed charge densities of the valence band maximum (VBM) and conduction band minimum (CBM), together with the planar-averaged electrostatic potential distribution along the out-of-plane (Z) direction, are also presented.

The electronic structures of monolayer CrSiCN₄ and CrGeCN₄ were evaluated using the HSE06 functional without spin-orbit coupling (SOC), as illustrated in **Figure 2**. Both monolayers exhibit direct band gap semiconducting behavior, with calculated band gaps of 1.46 eV and 1.19 eV for CrSiCN₄ and CrGeCN₄, respectively. The valence band maximum (VBM) and conduction band minimum (CBM) are located at the symmetry equivalent K and K' valleys, revealing the intrinsic valley characteristics of these systems. The surface electronic properties were further elucidated through the planar-averaged electrostatic potential profiles (**Figure 2(a, b)**). The work functions, obtained from the energy separation between the vacuum level and the Fermi level, are calculated to be 7.76 eV and 7.87 eV for CrSiCN₄ and CrGeCN₄, respectively. These large work functions indicate strong surface electron binding and suggest favorable interfacial compatibility with appropriate electrode materials, which is crucial for controlling carrier injection and band alignment in emerging valleytronic, optoelectronic, and photocatalytic applications. Notably, the

electrostatic potential profiles display a clear asymmetry between the two surfaces, originating from the intrinsic out-of-plane polarization. This polarization-induced dipole generates an internal electric field across the monolayers and leads to unequal vacuum potentials on opposite surfaces. The resulting surface potential differences are calculated as $\Delta\Phi = 1.45$ eV for CrSiCN₄ and 0.90 eV for CrGeCN₄, respectively, confirming the presence of a strong polarization-driven electrostatic imbalance.

The spin-orbit-coupled (SOC) induced band structures were further analyzed to explore the valleytronic properties, as shown in **Figure 3**. The SOC-inclusive band structures reveal reduced band gaps of 1.23 eV and 1.09 eV for CrSiCN₄ and CrGeCN₄, respectively, calculated using the PBE functional. These values are slightly smaller than the corresponding HSE06 results, consistent with the well-known tendency of the PBE functional to underestimate semiconductor band gaps due to the absence of exact exchange contributions. Despite this quantitative deviation, the PBE+SOC calculations successfully reproduce the essential electronic features and preserve the fundamental band characteristics of both monolayers. Therefore, considering the substantial computational cost of SOC calculations, the PBE functional was employed for subsequent SOC-dependent analyses, providing an effective balance between computational efficiency and reliable description of the relevant electronic properties.

The spin-polarized band structures and projected density of states (PDOS) without SOC are provided in the Appendix. The complete overlap between spin-up and spin-down channels confirms the nonmagnetic ground state of both monolayers. Upon inclusion of SOC, the semiconducting nature is preserved, while the spin degeneracy of the valley-edge states is lifted. As shown in **Figure 3**, the spin expectation values reveal that the band-edge states at the K and K' valleys are dominated by the out-of-plane spin component (S_z), with negligible in-plane contributions. The K valley states exhibit predominantly spin-down polarization ($\langle S_z \rangle \approx -0.5$), whereas the time-reversal related K' valley states possess opposite spin polarization ($\langle S_z \rangle \approx +0.5$). The nearly degenerate K/K' valley energies are consistent with preserved time-reversal symmetry in these nonmagnetic systems and this small deviation can be associated with the moderate SOC strength of Cr-derived states compared with heavier transition-metal systems. Nevertheless, strong SOC-induced spin splitting occurs at the band edges, with valence-band splittings (Δ_v) of 53.3 meV and 52.4 meV for CrSiCN₄ and CrGeCN₄, respectively, while the conduction-band splitting (Δ_c) remains approximately 1.6 meV. The opposite spin polarization between the K and K' valleys establish robust spin-valley locking, providing the basis for valley-selective optical transitions and Berry-curvature driven valley Hall phenomena. Both optimized CrSiCN₄ and CrGeCN₄ monolayers are non-centrosymmetric, although their upper and lower sublayers exhibit an approximate horizontal mirror relationship. The absence of spatial inversion symmetry originates from the inequivalent atomic configurations, where inversion operation does not map each atomic site onto an equivalent site occupied by the same element. Together with SOC and preserved time-reversal symmetry, this broken inversion symmetry enables spin splitting of the valley-edge states.



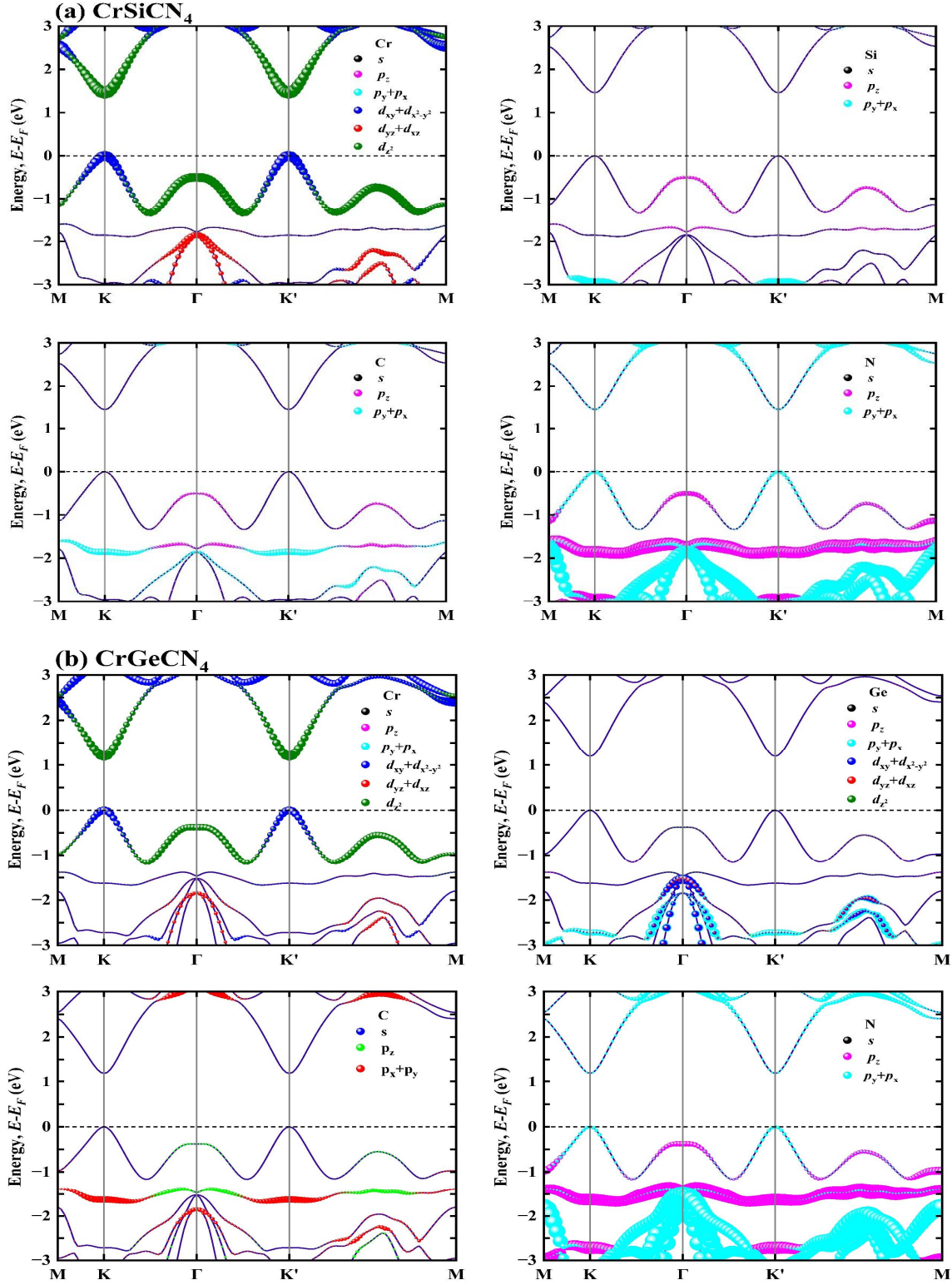


Figure 4. Orbital-projected band structures of monolayer (a)CrSiCN₄ and (b)CrGeCN₄, illustrating the orbital contributions of Cr-d, Si/Ge-p, C-p, and N-p states near the band edges.

The orbital-resolved projected density of states (PDOS) was further analyzed to elucidate the orbital contributions governing the electronic structures of CrSiCN₄ and CrGeCN₄, as presented in **Figure 2**. The PDOS profiles reveal that the band-edge states of both monolayers are primarily derived from Cr-*d* orbitals with significant hybridization from N-*p* states, whereas the Si and C derived states contribute minimally near the band edge. This indicates that the electronic properties around the Fermi level and the low-energy excitations of both systems are mainly governed by Cr-N orbital hybridization.

Further insight into the orbital character of the band-edge states is provided by the orbital-projected band structures in **Figure 4(a, b)**. The Si/Ge- and C-derived states contribute only weakly near the band edges, whereas pronounced Cr-*d* and N-*p* contributions are observed around the K/K' valleys. Specifically, the VBM is predominantly derived from the in-plane Cr-*d*_{xy} and *d*_{x²-y²} orbitals, strongly hybridized with N-*p*_x/*p*_y and N-*p*_z states, while the CBM is mainly characterized by Cr-*d*_{z²} states with appreciable N-*p*_x/*p*_y hybridization. This distinct orbital composition establishes Cr-N hybridization as the microscopic origin of the valley-edge states and underpins the SOC-driven spin-valley characteristics of both monolayers.

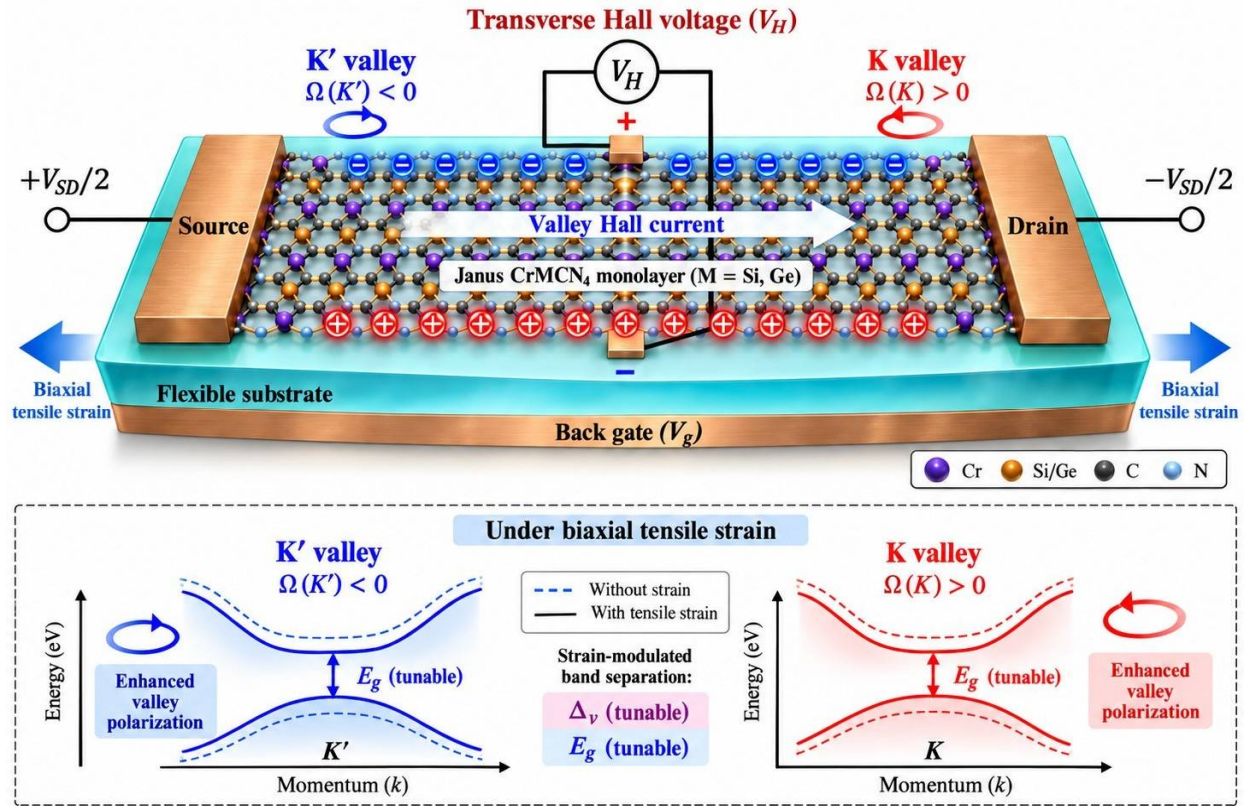


Figure 5. Schematic illustration of a strain-tunable spin-valley / valley Hall device based on Janus CrMCN₄ (M = Si, Ge). The monolayer channel is integrated on a flexible substrate between source and drain electrodes with back gate. Biaxial tensile strain modulates the electronic structure and enhances transverse Hall-voltage response. Charge accumulation at opposite channel edges produce a measurable Hall voltage V_H .

3.3.1 Comparative Analysis

To assess the valleytronic characteristics of monolayer CrSiCN₄ and CrGeCN₄, the SOC-induced spin splittings at the band edges are benchmarked against representative two-dimensional valleytronic materials, as summarized in [Table 2](#). Here, Δ_v and Δ_c denotes the spin splitting magnitudes at the valence and conduction band edges, respectively. Both monolayers exhibit pronounced band-edge spin-splitting asymmetry, with valence-band splittings of 53.3 meV and 52.4 meV for CrSiCN₄ and CrGeCN₄, respectively, accompanied by much smaller conduction-band splittings of 1.6 meV. The obtained valence-band splittings are comparable to several reported valleytronic systems, including CrC₂N₄, VSi₂P₄, VBNS₂, and WS₂/MnO₂, while remaining lower than the larger splittings reported for CrSi₂N₄, MoSi₂N₄, and CrSi₂P₄, which originate from their stronger SOC interactions.

Table 2. Comparison of valley related band-edge splitting in CrSiCN₄, CrGeCN₄, and selected representative 2D valleytronic systems. The values Δ_v and Δ_c are reported in meV. For nonmagnetic systems, these quantities represent SOC-induced valley-edge spin splitting, whereas for magnetic or proximity-coupled systems, they correspond to K/K' valley splitting or valley polarization.

Compound	Δ_v (meV)	Δ_c (meV)	Reference
CrSiCN ₄	53.3	1.6	This work
CrGeCN ₄	52.4	1.6	This work
CrC ₂ N ₄	51.9	1.7	[24]
CrSi ₂ N ₄	130	26	[22]
MoSi ₂ N ₄	130-172	3-17	[16]
CrSi ₂ P ₄	170	23	[22]
VSi ₂ P ₄	49.4	3	[35]
VBNS ₂	48.6	-	[36]
WS ₂ /MnO ₂	43	-	[37]

The pronounced Δ_v/Δ_c asymmetry arises from the orbital-dependent SOC response of the band-edge states. The VBM dominated by Cr- $d_{xy}/d_{x^2-y^2}$ orbitals strongly hybridized with N- p_{xy} states which exhibits enhanced SOC-induced splitting, whereas the Cr- d_{z^2} dominated CBM shows comparatively weaker splitting. Therefore, CrSiCN₄ and CrGeCN₄ represent promising nonmagnetic valleytronic platforms by combining substantial band-edge spin splitting, robust spin-valley locking, and valley-contrasting Berry curvature, highlighting their suitability for valley-dependent electronic and optoelectronic applications. Building on these intrinsic properties, [Figure 5](#) illustrates a prototype strain-controlled valley Hall device concept for tuning valley-dependent transport.

3.4 Valley Physics: Berry Curvature

The Berry curvature serves as a momentum-space geometric descriptor of electronic Bloch states and acts as an effective magnetic field in reciprocal space, governing anomalous carrier dynamics and valley-dependent transport phenomena. In time-reversal-symmetric systems, the Berry curvature at the K and K' valleys exhibit opposite signs, resulting in valley-contrasting electronic responses. Following the valley effective model proposed by Li *et al.* [16], the low-energy electronic states near the inequivalent K/K' valleys can be described within a two-band Dirac Hamiltonian framework. Without considering spin-orbit coupling (SOC), the Hamiltonian around the valley index $\tau = \pm 1$ is expressed as:

$$H_0^\tau = \alpha(\tau k_x \sigma_x + k_y \sigma_y) + \frac{\Delta}{2} \sigma_z, \quad (2)$$

Here, k_i denotes the wave vector measured relative to the valley center, σ_i represents the Pauli matrices in the two-orbital basis, α is the effective coupling coefficient and Δ corresponds to the band-gap parameter. Upon incorporating spin-orbit coupling (SOC), the effective Hamiltonian is modified as:

$$H^\tau = \alpha(\tau k_x \sigma_x + k_y \sigma_y) + \frac{\Delta}{2} \sigma_z - \frac{\lambda}{2} \tau (\sigma_z - 1) \hat{s}_z, \quad (3)$$

Here, λ represents the effective SOC strength, and $\hat{s}_z$ denotes the spin Pauli matrices. This Hamiltonian captures the essential spin-valley-coupled nature of the band-edge states, where SOC induces pronounce splitting in the valence band while producing comparatively weaker splitting in the conduction band. For Berry curvature, the following scheme can be used:

$$\Omega_{n,z}(\mathbf{k}) = -2 \text{Im} \sum_{m \neq n} \frac{\langle n\mathbf{k} | \hat{v}_x | m\mathbf{k} \rangle \langle m\mathbf{k} | \hat{v}_y | n\mathbf{k} \rangle}{(\omega_m - \omega_n)^2}, \quad (4)$$

In this expression, $\hat{v}_x$ and $\hat{v}_y$ correspond to the velocity operators along the in-plane directions, while $E_n = \hbar \omega_n$ denotes the eigenvalue associated with the Bloch state $|n\mathbf{k}\rangle$. These velocity operators are evaluated directly from the effective Hamiltonian through:

$$\hat{v}_i = \frac{1}{\hbar} \frac{\partial H^\tau}{\partial k_i}, \quad i = x, y \quad (5)$$

Incorporating the SOC contribution into the effective Hamiltonian (Equation 2), the Berry curvature associated with the conduction band for a given valley index τ and spin state $s = \pm 1$ can be expressed as [16]:

$$\Omega_c^\tau(\mathbf{k}, s) = -\tau \frac{2\alpha^2 \tilde{\Delta}_{\tau s}}{(\tilde{\Delta}_{\tau s}^2 + 4\alpha^2 k^2)^{3/2}}, \quad (6)$$

here

$$\tilde{\Delta}_{\tau s} = \Delta - \tau s \lambda, \quad (7)$$

For the corresponding spin-split valence band, the Berry curvature exhibits an opposite sign,

$$\Omega_v^\tau(\mathbf{k}, s) = -\Omega_c^\tau(\mathbf{k}, s). \quad (8)$$

Equation 5 demonstrates that the Berry curvature reverses its sign between the two time-reversal related K and K' valleys due to the valley dependent factor τ . Consequently, the K and K' valleys exhibit finite Berry curvatures of opposite signs, with the maximum magnitude occurring at the valley center ($k=0$).

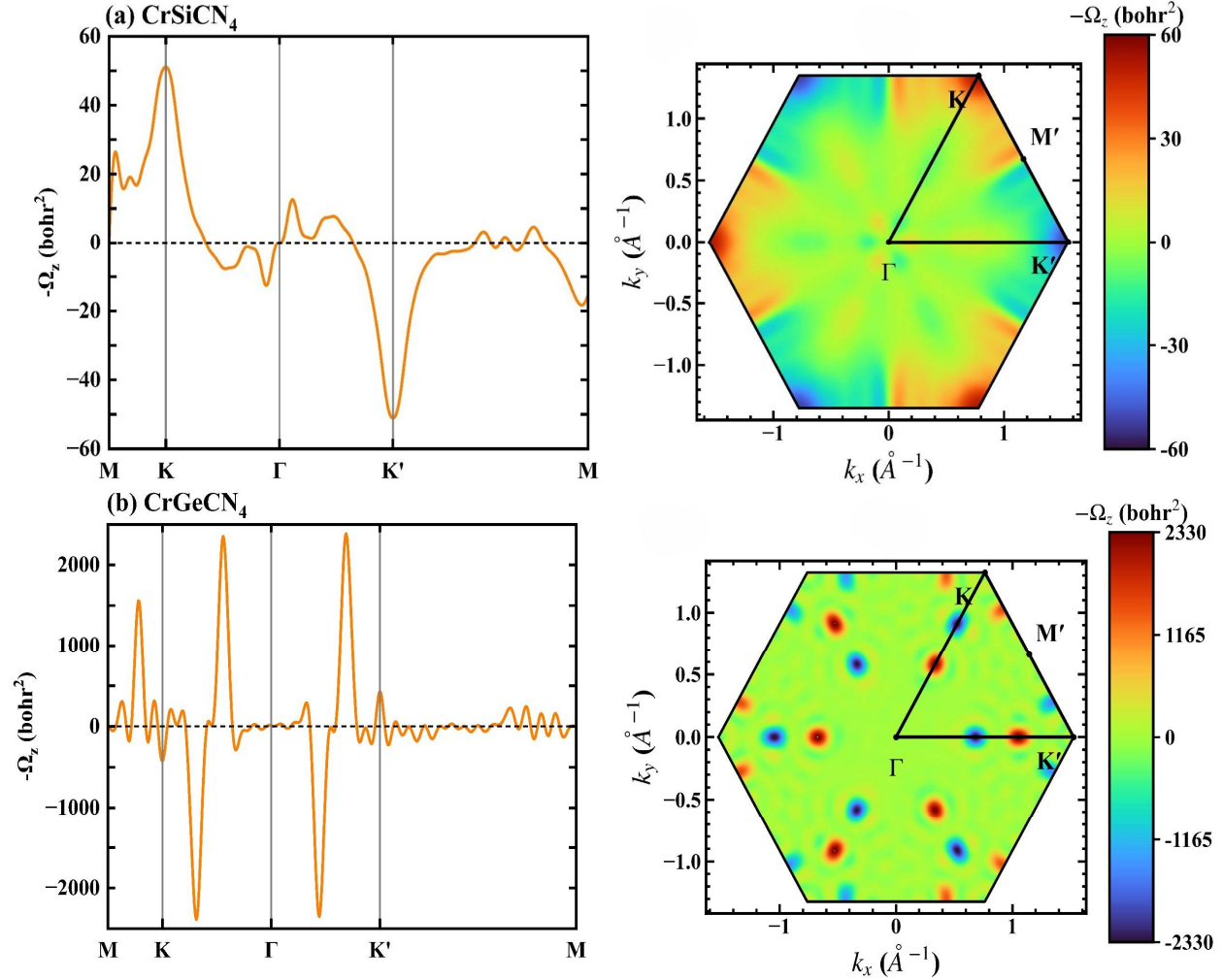


Figure 6. Berry curvature, $-\Omega_z(\text{bohr}^2)$, of monolayer (a) CrSiCN₄ and (b) CrGeCN₄ along the high-symmetry path $M \rightarrow K \rightarrow \Gamma \rightarrow K' \rightarrow M$, showing opposite Berry curvature at the K and K' valleys. The corresponding contour maps reveal the valley-contrasting Berry curvature distribution throughout the hexagonal Brillouin zone.

Under an applied in-plane electric field, the finite Berry curvature induces an anomalous transverse velocity, giving rise to Hall-type transport phenomena such as the valley Hall effect. Since CrSiCN₄ and CrGeCN₄ preserve time-reversal symmetry, the net charge Hall conductivity vanishes when the K and K' valleys are equally occupied. However, the opposite Berry curvatures at these time-reversal related valleys drive carriers toward opposite transverse directions, resulting in a pure valley Hall current. This valley-contrasting response can be expressed as:

$$\Omega_z(K) = -\Omega_z(K') \quad (9)$$

Therefore, despite the absence of a net charge Hall current, the opposite anomalous velocities generated in the K and K' valleys produce a finite valley-polarized Hall response.

The Berry curvature of monolayer CrSiCN₄ and CrGeCN₄ exhibits strong valley-contrasting behavior, satisfying $\Omega_z(K) = -\Omega_z(K')$ due to time-reversal symmetry. This opposite Berry curvature at the time-reversal-related valleys produces anomalous transverse carrier motion under an in-plane electric field, described by $v_a = -\frac{e}{h} \mathbf{E} \times \boldsymbol{\Omega}(\mathbf{k})$, giving rise to valley-dependent Hall transport. Although the charge Hall conductivity cancels when the K and K' valleys are equally populated, the opposite anomalous velocities of carriers from the two valleys generate a finite valley Hall response. Such valley-contrasting Berry curvature and valley Hall physics have been widely established in inversion-asymmetric two-dimensional systems [4,38,39].

For CrSiCN₄, the Berry curvature is mainly concentrated around the K/K' valleys with opposite signs, reflecting the intrinsic valley-dependent electronic structure. In contrast, CrGeCN₄ exhibits a strongly enhanced Berry-curvature landscape with sharp hot spots reaching the order of 10³ bohr², significantly exceeding the response of CrSiCN₄. This enhancement cannot be attributed solely to stronger valley-edge SOC because both systems exhibit nearly identical SOC-induced spin splittings. Furthermore, the reduction of the fundamental band gap provides only a limited enhancement within the massive Dirac picture. Instead, the giant Berry curvature in CrGeCN₄ is attributed to multiband electronic reconstruction induced by Ge substitution, including modified orbital hybridization, band alignment, and reduced interband energy separation.

According to the Kubo formulation, Berry curvature depends on interband velocity matrix elements and the inverse square of energy separation (Equation 4). Therefore, small energy separations or SOC-mediated avoided crossings can strongly amplify the Berry-curvature response, as shown in Figure 3(b) near Γ point. Similar interband-driven enhancement of Berry curvature and Hall-related phenomena has been reported in multiband electronic systems [40–42]. Hence, the unusually large Berry-curvature hot spots in CrGeCN₄ originate primarily from enhanced multiband coupling rather than a simple increase of intrinsic valley curvature.

3.5 Photocatalytic Properties

Among the two monolayers, CrSiCN₄ exhibits a band gap exceeding the thermodynamic minimum of 1.23 eV required for overall water splitting. Its photocatalytic activity was assessed by aligning the band edges with the water redox potentials, where the CBM and VBM must provide sufficient driving forces for HER and OER, respectively.

Owing to its Janus asymmetry, CrSiCN₄ exhibits an intrinsic electrostatic potential difference of 1.48 eV, generating a built-in electric field from the lower to the upper surface (Figure 7). This field promotes spatial separation of photogenerated carriers, driving electrons toward the lower surface for HER and holes toward the upper surface for OER [43]. Accordingly, the H⁺/H₂ and O₂/H₂O redox potentials were referenced to the corresponding surface-specific vacuum levels. The redox potentials of water were referenced to the vacuum level using the following relations [44]:

$$V_{H^+/H_2} = E_{\text{vacuum}} - 4.44 + \text{pH} \times 0.059 \text{ eV} \quad (10)$$

$$V_{\text{O}_2/\text{H}_2\text{O}} = E_{\text{vacuum}} - 5.67 + \text{pH} \times 0.059 \text{ eV} \quad (11)$$

The band-edge alignment satisfies the OER requirement but provides insufficient driving force for HER, indicating that further band-edge modulation is necessary. Strain engineering therefore offers a promising route to optimize the band alignment and enable overall water splitting.

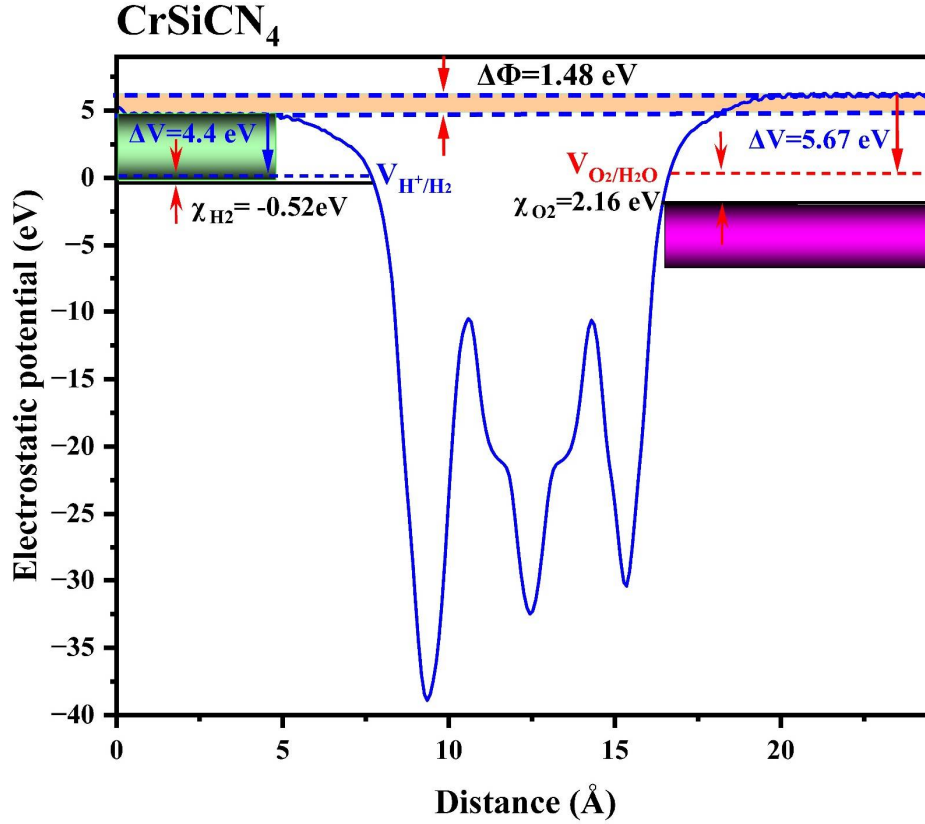


Figure 7. Potential profiles. The blue and red dashed lines represent the H_2 reduction and H_2O oxidation potentials, while the green and magenta regions indicate the CBM and VBM positions, respectively. $\chi(\text{H}_2)$ and $\chi(\text{O}_2)$ denote the HER and OER driving forces. $\Delta\Phi$ refers to the vacuum-level difference between the two surfaces.

3.6 Piezoelectric Properties

The piezoelectric behavior of both Janus monolayers is further examined by considering their in-plane and out-of-plane responses. Owing to the $P3m1$ symmetry for 2D, the elastic and piezoelectric tensors can be expressed in the following reduced forms [45,46]:

$$\mathbf{C} = \begin{pmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{11} & 0 \\ 0 & 0 & \frac{C_{11} - C_{12}}{2} \end{pmatrix}, \quad (12)$$

$$\mathbf{e} = \begin{pmatrix} e_{11} & -e_{11} & 0 \\ 0 & 0 & -e_{11} \\ e_{31} & e_{31} & 0 \end{pmatrix}, \quad (13)$$

$$\mathbf{d} = \begin{pmatrix} d_{11} & -d_{11} & 0 \\ 0 & 0 & -2d_{11} \\ d_{31} & d_{31} & 0 \end{pmatrix}. \quad (14)$$

The independent d_{11} and d_{31} coefficients can be derived from the relation $e_{ij} = d_{ij}C_{ij}$ as follows:

$$d_{11} = \frac{e_{11}}{C_{11} - C_{12}} \text{ and } d_{31} = \frac{e_{31}}{C_{11} + C_{12}}. \quad (15)$$

As HSE and PBE yield similar electronic characteristics, apart from band-gap differences, the PBE functional was adopted for computational efficiency. Using the orthorhombic supercell (**Figure 1**), the piezoelectric coefficients were calculated from **Equation (12)-(15)** and are summarized in **Table 3**. The broken inversion symmetry of Janus CrSiCN₄ and CrGeCN₄ gives rise to both in-plane and out-of-plane piezoelectric responses. As shown in **Figure 8**, the calculated e_{11} values are 11.157 and $13.136 \times 10^{-10} \text{ C m}^{-1}$ for CrSiCN₄ and CrGeCN₄, respectively, with the electronic contribution dominating the in-plane response. These values are considerably larger than those reported for MoBNTe₂ ($3.008 \times 10^{-10} \text{ C m}^{-1}$), MoSiGeN₄ ($5.116 \times 10^{-10} \text{ C m}^{-1}$), and MoSiN₃H ($4.146 \times 10^{-10} \text{ C m}^{-1}$), highlighting the pronounced in-plane piezoelectricity of the present monolayers. In contrast, the electronic and ionic contributions to e_{31} have opposite signs and partially compensate each other, resulting in comparatively weaker out-of-plane responses.

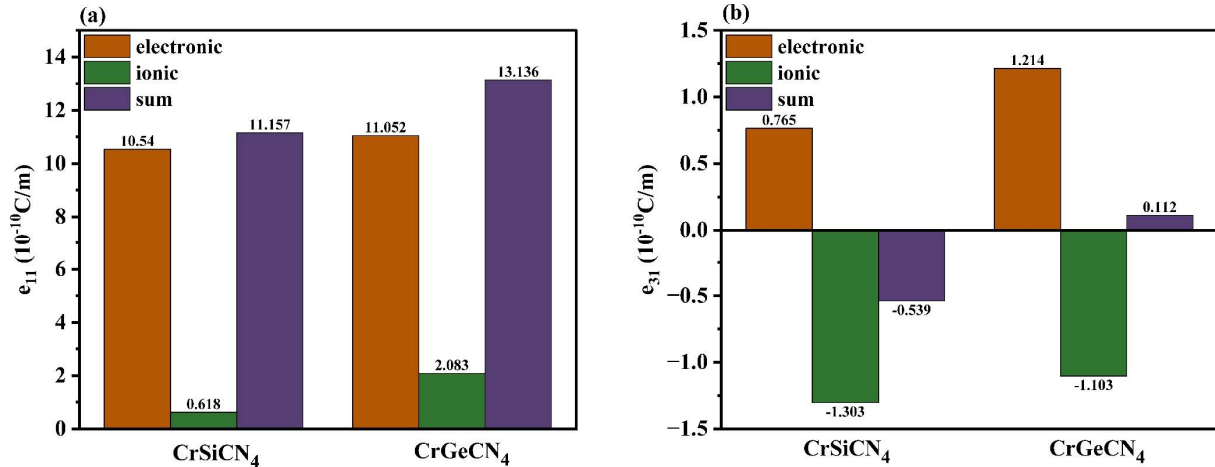


Figure 8. Piezoelectric stress coefficients (a) e_{11} and (b) e_{31} of Janus CrSiCN₄ and CrGeCN₄ monolayers, showing the electronic, ionic, and total contributions.

The corresponding piezoelectric strain coefficients d_{11} are 3.305 and 5.084 pm V^{-1} for CrSiCN₄ and CrGeCN₄, respectively. This demonstrates the stronger in-plane electromechanical response of CrGeCN₄. Conversely, CrSiCN₄ exhibits a larger d_{31} of 0.082 pm V^{-1} compared with 0.020 pm V^{-1} for CrGeCN₄. Notably, the d_{31} value of CrSiCN₄ exceeds those reported for MoBNTe₂ (0.062 pm V^{-1}), MoSiGeN₄ (0.014 pm V^{-1}), and MoSiN₃H (0.058 pm V^{-1}), indicating a competitive

vertical piezoelectric response. Overall, the large in-plane response of CrGeCN₄ and enhanced vertical response of CrSiCN₄ demonstrate complementary piezoelectric characteristics, making these Janus monolayers promising candidates for nanoscale electromechanical and energy-harvesting applications.

Table 3. Piezoelectric coefficients e_{11}/e_{31} (10^{-10} C m⁻¹) and d_{11}/d_{31} (pm V⁻¹) of CrMCN₄ (M = Si/Ge).

Material	e_{11}	d_{11}	e_{31}	d_{31}	Reference
CrSiCN ₄	11.157	3.305	0.539	0.082	This work
CrGeCN ₄	13.136	5.084	0.112	0.020	This work
MoBNTe ₂	3.008	4.754	0.188	0.062	[47]
MoSiGeN ₄	5.116	1.494	0.087	0.014	[19]
MoSiN ₃ H	4.146	1.622	0.290	0.058	[48]

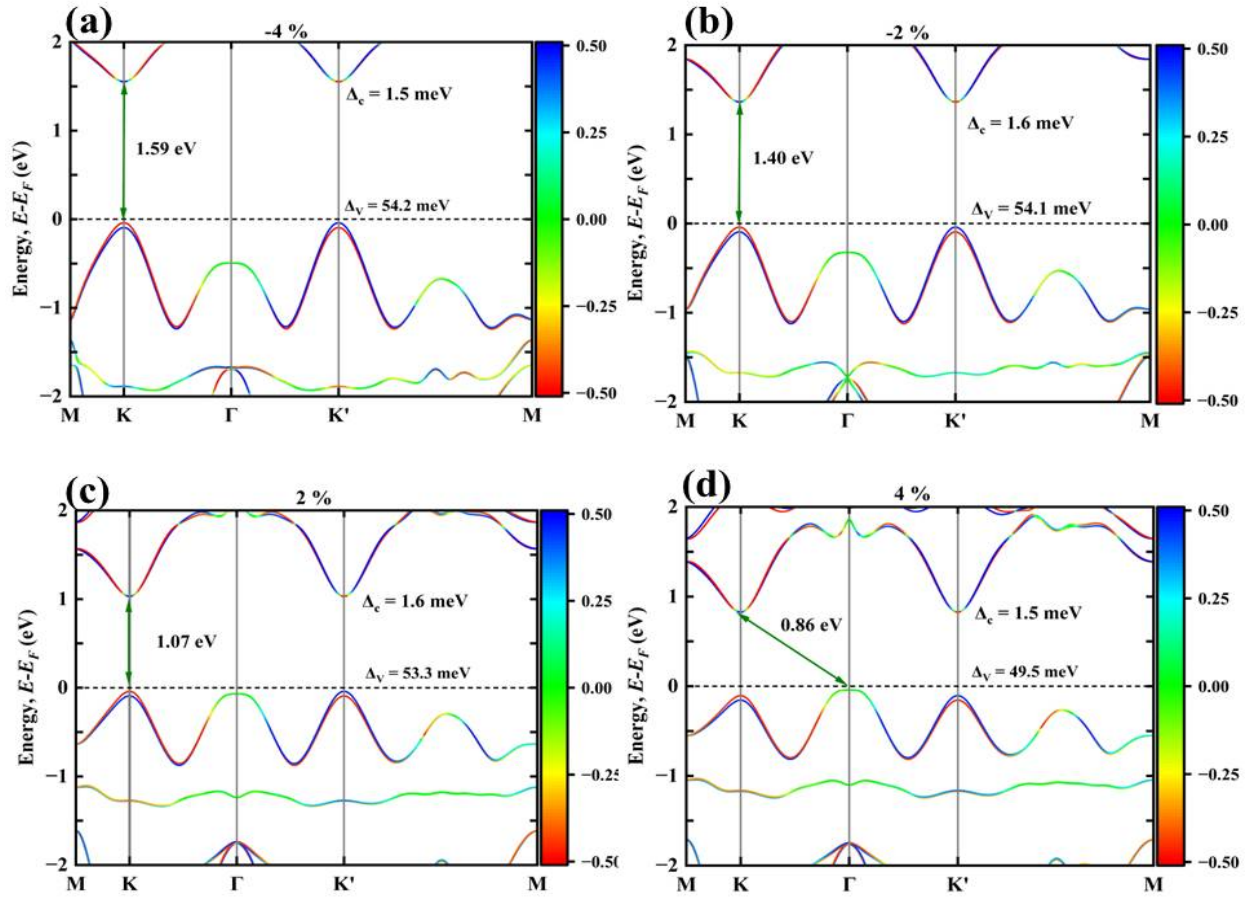


Figure 9. Spin-resolved band structures of monolayer CrSiCN₄ with SOC under biaxial strains of (a) -4%, (b) -2%, (c) +2%, and (d) +4%. The color scale represents the out-of-plane spin expectation value $\langle S_z \rangle$, indicating strain-induced spin splitting and valley polarization.

3.7 Effect of Biaxial Strain on Band Structure, Spin Splitting

The strain-dependent electronic and spin properties of monolayer CrSiCN₄ and CrGeCN₄ were systematically explored using SOC inclusive spin-resolved band structure calculations. The results reveal that biaxial strain provides an effective means to simultaneously modulate the band topology, valley-dependent spin characteristics, and carrier transport properties. As illustrated in **Figure 9(a–d)** and **Figure 10(a–d)**, lattice deformation induces substantial shifts of the conduction-band minimum (CBM) and valence-band maximum (VBM), resulting in pronounced strain-driven electronic reconstruction.

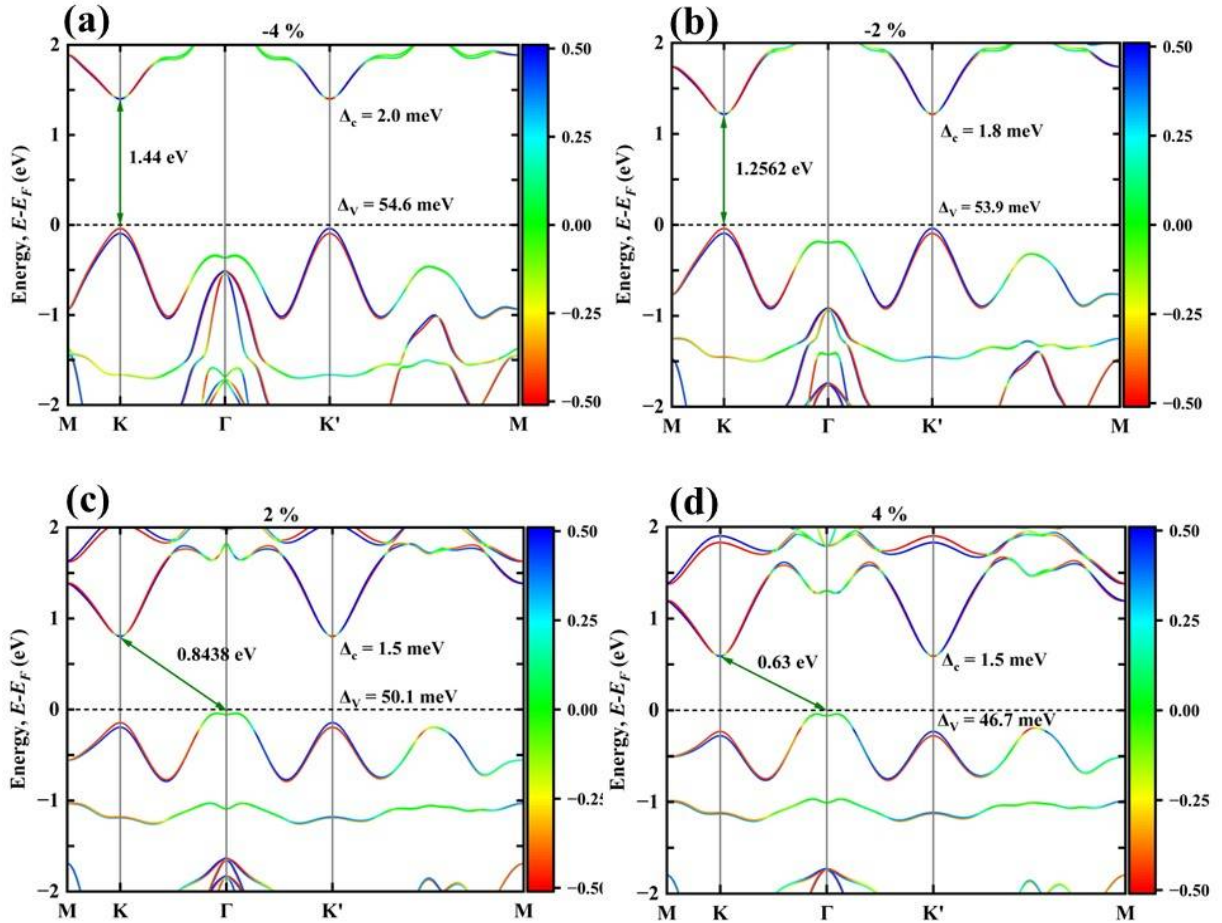


Figure 10. Spin-resolved band structures of monolayer CrGeCN₄ with SOC under biaxial strains of (a) -4%, (b) -2%, (c) +2%, and (d) +4%. The color scale represents the out-of-plane spin expectation value $\langle S_z \rangle$, indicating strain-induced spin splitting and valley polarization.

In the absence of strain, CrSiCN₄ exhibits a direct band gap of 1.23 eV at the K valley, whereas CrGeCN₄ possesses a smaller direct gap of 1.09 eV. Compressive strain enhances the band separation, increasing the gaps to 1.59 eV and 1.44 eV at -4% strain for CrSiCN₄ and CrGeCN₄, respectively. In contrast, tensile strain progressively narrows the gaps, reaching 0.86 eV and 0.63 eV at +4% strain. This monotonic reduction originates from the strain-induced modification of orbital hybridization and crystal-field splitting, which alters the relative energetic positions of the

CBM and VBM. Notably, tensile strain beyond +2% drives a direct-to-indirect band-gap transition in both systems, providing an effective route for strain-controlled optical and electronic functionalities.

The inclusion of SOC reveals pronounced spin splitting near the valley edges with opposite spin orientations at the time-reversal related K and K' valleys, demonstrating robust valley-contrasting spin polarization. The conduction-band splitting remains relatively weak and nearly strain independent, with Δ_c values of 1.5-1.6 meV for CrSiCN₄ and 1.5-2.0 meV for CrGeCN₄. In contrast, the valence-band splitting reaches considerably larger values of 49.5-54.2 meV and 46.7-54.6 meV, respectively. The pronounced asymmetry between Δ_v and Δ_c indicates a stronger SOC contribution to the valence band states, while the nearly invariant splitting under strain suggests that the intrinsic spin-valley coupling is preserved against moderate lattice deformation. The comparable spin-splitting magnitudes in both compounds indicate that the Cr-based electronic framework plays the dominant role, while Ge substitution mainly provides additional chemical modulation through its larger atomic size and stronger atomic SOC contribution.

Beyond the modulation of electronic and spin characteristics, biaxial strain provides an effective pathway for engineering the carrier dynamics through the reconstruction of band-edge dispersion. The calculated electron and hole effective masses along the K- Γ and K-M directions (**Figure 11(c, d)**) reveal pronounced anisotropic transport characteristics in both CrSiCN₄ and CrGeCN₄. The electron effective mass exhibits a systematic reduction along the K- Γ direction under deformation, reflecting enhanced band curvature and increased electronic dispersion. For instance, in CrSiCN₄, the electron effective mass decreases from 1.51 m_0 at -4% strain to 0.87 m_0 at +4% strain, accompanied by a similar evolution in CrGeCN₄. In contrast, the hole effective masses display a more non-monotonic and direction dependent response, highlighting the distinct sensitivity of valence band states to lattice deformation. This contrasting behavior between electron and hole carriers demonstrates that strain does not merely tune the magnitude of the band gap but actively reshapes the momentum-dependent electronic landscape governing carrier transport. The observed anisotropic effective-mass evolution originates from strain-induced modifications of valley dispersion and band-edge curvature, providing an additional degree of freedom for directional carrier control. Therefore, strain engineering in CrSiCN₄ and CrGeCN₄ offers a versatile strategy to tailor electronic transport characteristics and complementing their robust spin-valley properties for multifunctional two-dimensional valleytronic applications.

Overall, these results establish strain as an efficient strategy for engineering the coupled electronic, spin, and transport properties of CrSiCN₄ and CrGeCN₄. The coexistence of a widely tunable band gap, robust valley-contrasting spin polarization with sizable valence-band splitting (~ 50 meV), and strain-dependent carrier anisotropy highlights these two-dimensional materials as promising platforms for strain-controlled valleytronic, and optoelectronic applications.

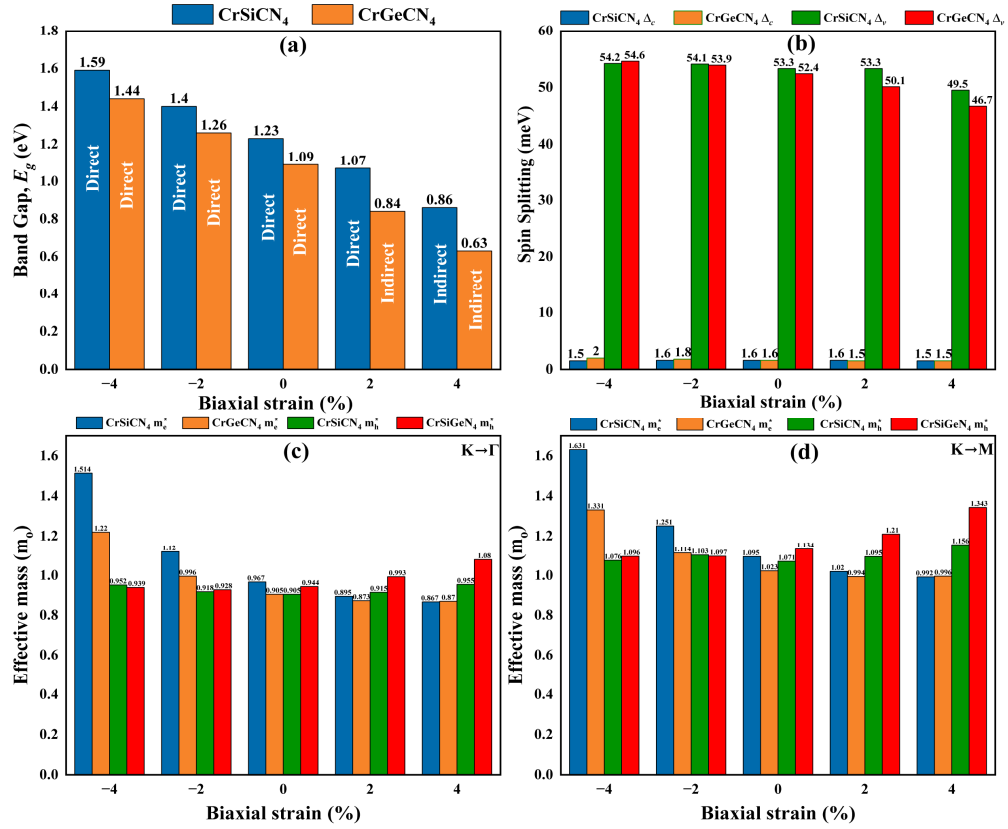


Figure 11. Effect of biaxial strain on the electronic and spin-dependent properties of monolayer CrSiCN₄ and CrGeCN₄. (a) Strain-induced modulation of the band gaps. (b) Variation of spin splitting at the conduction and valence band edges. (c, d) Evolution of carrier effective masses along the K-Γ and K-M directions, respectively, under different strain conditions.

3.8 Strain-dependent optical absorption

Mechanical strain is an established route for tuning the electronic and optical energy scales of two-dimensional semiconductors: tensile strain lowers the optical gap of MoS₂ [49], while first-principles calculations show that biaxial strain can strongly modify the band gaps of related MX₂ monolayers [50]. Strong visible-light absorption has also been reported for Janus MoSSe [51], PtSSe [52], and group-III chalcogenide monolayers [5]. **Figure 12** compares the calculated optical absorption spectra of CrSiCN₄ and CrGeCN₄ under biaxial strains from -4% to +4%. At equilibrium, CrSiCN₄ exhibits a pronounced low-energy peak at 2.92 eV with an absorption of 18.86%, followed by its dominant maximum at 9.66 eV (25.54%). The corresponding CrGeCN₄ features occur at lower energies, 2.64 eV (18.01%) and 9.16 eV (24.26%), respectively. Thus,

relative to CrSiCN₄, CrGeCN₄ shows a consistent displacement of the main spectral features toward lower photon energy without a uniform increase in peak magnitude.

The lower-energy response is particularly sensitive to strain. The first pronounced peak of CrSiCN₄ shifts from 3.24 eV at -4% to 2.48 eV at +4%, whereas the corresponding feature in CrGeCN₄ moves from 2.84 to 2.24 eV. Within the shaded visible interval (~ 1.65 -3.10 eV), the mean optical absorption increases from 7.92% to 11.20% for CrSiCN₄ and from 10.16% to 11.88% for CrGeCN₄ as the strain changes from -4% to +4%. Broad extension of light harvesting toward lower photon energies has been reported in other two-dimensional systems [53], while structural modification can also redshift prominent optical features [54]. In closely related Janus XMSiY₂ monolayers, tensile strain was reported to redshift the absorption onset, whereas compressive strain shifted it toward higher energies [55], in qualitative agreement with the strain-dependent optical response observed in the present CrMCN₄ monolayers.

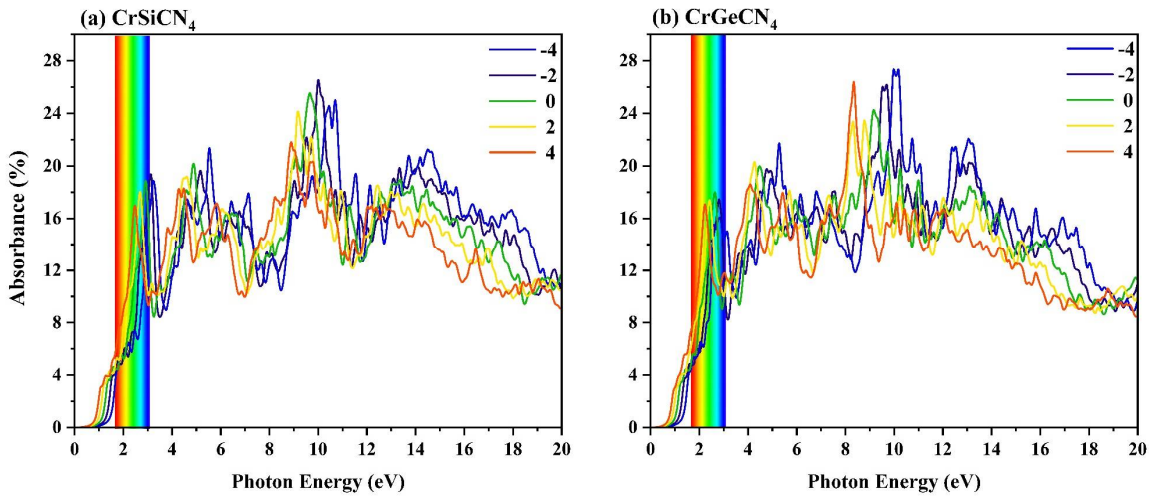


Figure 12. Strain-dependent optical absorption spectra of (a) CrSiCN₄ and (b) CrGeCN₄ monolayers under biaxial strains of -4%, -2%, 0%, +2%, and +4%. The shaded region denotes the visible-light range.

Strain redistributes the higher-energy part of the spectra rather than simply scaling the curves. The robust picture is therefore a strain-controlled migration of the dominant absorption features, accompanied by material-dependent intensity redistribution. The spectra for CrGeCN₄ remains generally shifted toward lower photon energy and shows the larger average response in the visible window, whereas that of CrSiCN₄ displays the slightly larger total shift of its dominant peak across the studied strain range. These trends indicate a tunable optical spectral window under moderate biaxial strain and may be relevant to strain-controlled optoelectronic design. The absorption spectra alone, however, do not establish specific orbital transitions, excitonic character, photovoltaic efficiency, photocurrent, or complete device performance.

4 Conclusions

Janus CrSiCN₄ and CrGeCN₄ monolayers combine spin-valley coupling with composition-dependent Berry curvature and complementary piezoelectric responses. HSE06 hybrid-functional calculations confirm that both unstrained monolayers are direct-gap semiconductors, with band gaps of 1.46 eV for CrSiCN₄ and 1.19 eV for CrGeCN₄. Cr-*d*/N-*p* hybridization dominates the frontier states, with orbital-dependent spin-orbit coupling producing valence-band spin splitting near 53 meV and conduction-band splitting near 1.6 meV. Opposite out-of-plane spin character and Berry curvature at K and K' provide the basis for valley-dependent transverse transport while preserving time-reversal symmetry. The much stronger Berry-curvature hot spots in CrGeCN₄, despite comparable valley-edge spin splittings, point to a role for substitution-sensitive multiband coupling.

Biaxial strain from -4% to +4% narrows the PBE+SOC gaps from 1.59 to 0.86 eV in CrSiCN₄ and from 1.44 to 0.63 eV in CrGeCN₄. Tensile strain induces indirect-gap behavior while retaining K/K' spin contrast, modifies carrier effective masses, and shifts absorption toward lower photon energies. Ge substitution increases the in-plane piezoelectric coefficient d_{11} from 3.305 to 5.084 pm V⁻¹, whereas CrSiCN₄ exhibits the larger out-of-plane magnitude, $|d_{31}| = 0.082$ pm V⁻¹, compared with 0.020 pm V⁻¹ for CrGeCN₄. The unstrained CrSiCN₄ band alignment remains insufficient for hydrogen evolution, making further band-edge engineering necessary for overall water splitting. These results provide a microscopic basis for selecting composition and strain to develop Cr-based Janus materials for valleytronic, optoelectronic, and electromechanical applications.

Data availability

The data sets generated and/or analyzed in this study are available from the corresponding author on reasonable request.

Declaration of interest

The authors declare no competing interests.

CRediT authorship contribution statement

Jubair Hossan Abir: Conceptualization, Software, Methodology, Formal analysis, Data curation, Visualization, Writing-original draft, review & editing; **Md Rabi Us Sani Shapon:** Software, Formal analysis, Data curation, Visualization, Writing-draft, review & editing; **Sarker Sukanta Babu Pallab & Tanvir Khan:** Formal analysis, Visualization, Writing- draft; **Raihana Shams Islam:** Conceptualization, Supervision, Validation, Administration, Writing- Reviewing and Editing; **Saleh Hasan Naqib:** Conceptualization, Supervision, Validation, Administration, Writing- Reviewing and Editing.

Appendix A

Wannier interpolation

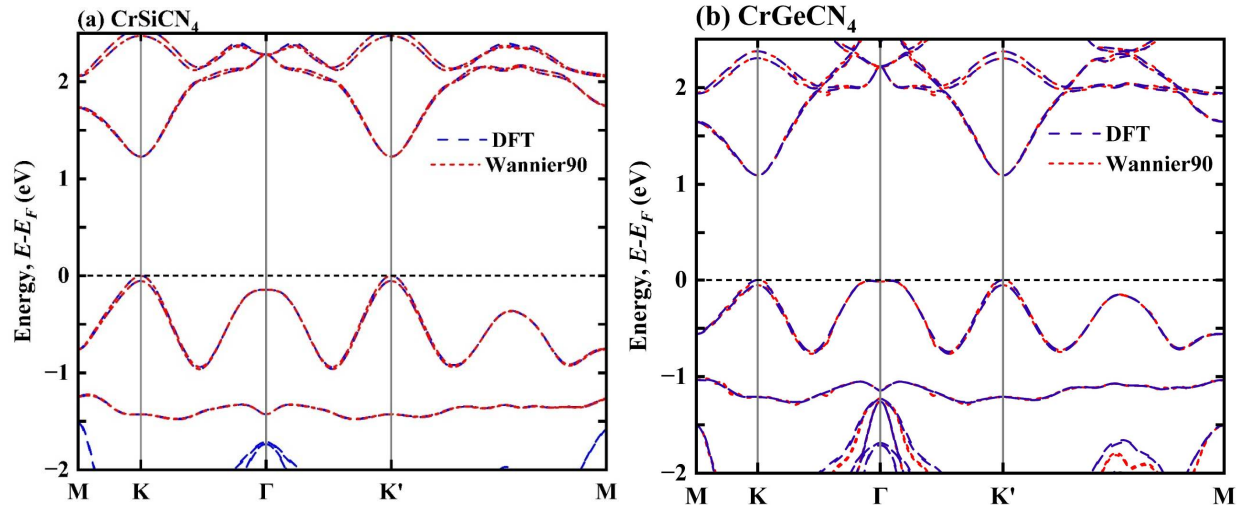


Figure 13. Figure shows a side-by-side comparison of electronic band structures derived from conventional DFT methods (dashed blue) and those reconstructed using maximally localized Wannier functions (dashed red)

Spin-Polarized Electronic Structure without SOC

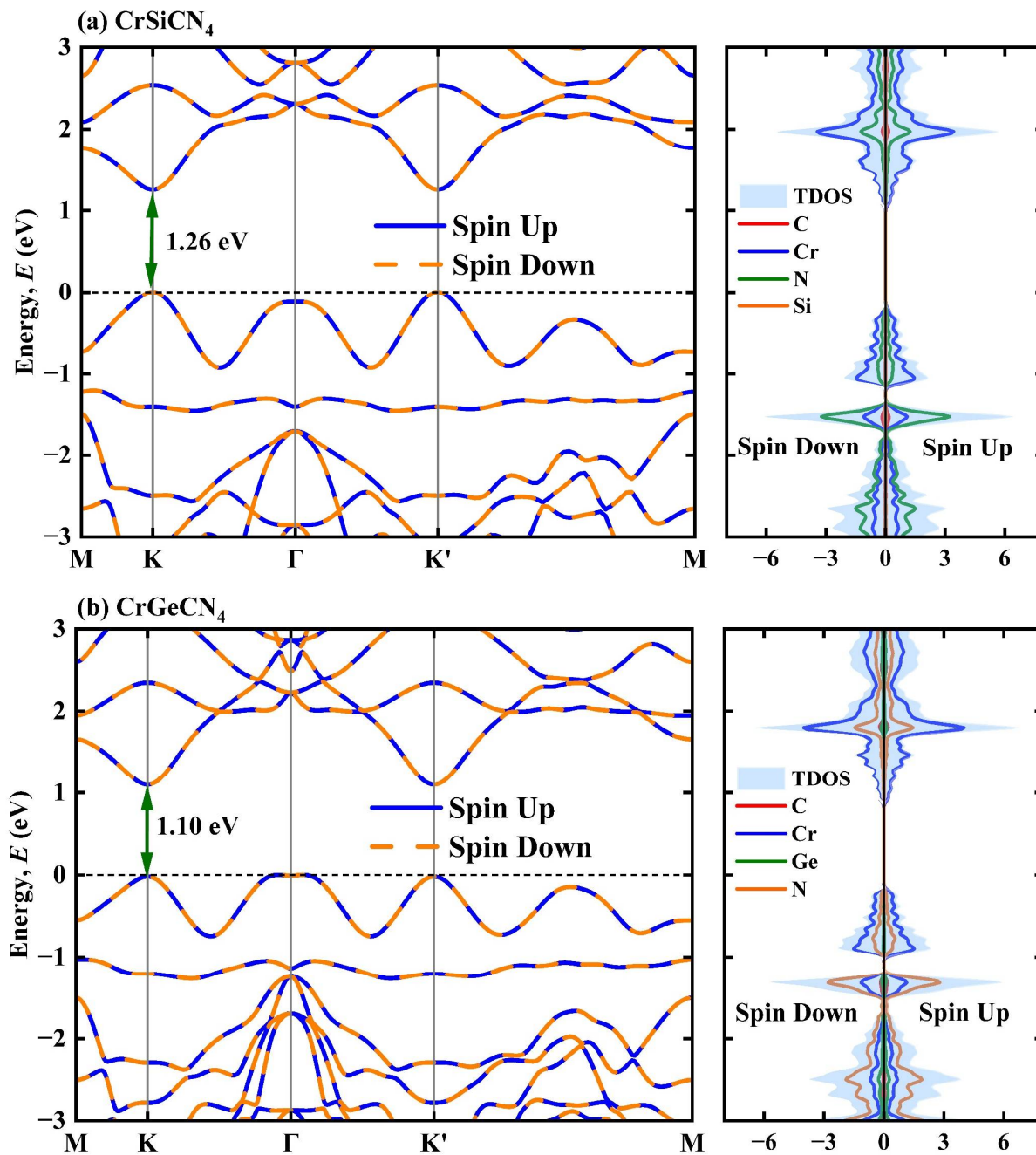


Figure 14. Spin-resolved band structure of monolayer (a) CrSiCN₄ and (b) CrGeCN₄. The right-side panel represents Spin-resolved projected density of states (PDOS). The Fermi level was set to 0 eV.

References

- [1] J.R. Schaibley, H. Yu, G. Clark, P. Rivera, J.S. Ross, K.L. Seyler, W. Yao, X. Xu, Valleytronics in 2D materials, *Nature Reviews Materials*, 1 (2016) 16055. <https://doi.org/10.1038/natrevmats.2016.55>.
- [2] Z. Peng, X. Chen, Y. Fan, D.J. Srolovitz, D. Lei, Strain engineering of 2D semiconductors and graphene: from strain fields to band-structure tuning and photonic applications, *Light: Science & Applications*, 9 (2020) 190. <https://doi.org/10.1038/s41377-020-00421-5>.
- [3] J. Du, H. Yu, B. Liu, M. Hong, Q. Liao, Z. Zhang, Y. Zhang, Strain Engineering in 2D Material-Based Flexible Optoelectronics, *Small Methods*, 5 (2021) 2000919. <https://doi.org/10.1002/smt.202000919>.
- [4] D. Xiao, G.-B. Liu, W. Feng, X. Xu, W. Yao, Coupled Spin and Valley Physics in Monolayers of MoS₂ and Other Group-VI Dichalcogenides, *Physical Review Letters*, 108 (2012) 196802. <https://doi.org/10.1103/PhysRevLett.108.196802>.
- [5] T. Cao, G. Wang, W. Han, H. Ye, C. Zhu, J. Shi, Q. Niu, P. Tan, E. Wang, B. Liu, J. Feng, Valley-selective circular dichroism of monolayer molybdenum disulphide, *Nature Communications*, 3 (2012) 887. <https://doi.org/10.1038/ncomms1882>.
- [6] K.F. Mak, K. He, J. Shan, T.F. Heinz, Control of valley polarization in monolayer MoS₂ by optical helicity, *Nature Nanotechnology*, 7 (2012) 494–498. <https://doi.org/10.1038/nnano.2012.96>.
- [7] Z. Wu, B.T. Zhou, X. Cai, P. Cheung, G.-B. Liu, M. Huang, J. Lin, T. Han, L. An, Y. Wang, S. Xu, G. Long, C. Cheng, K.T. Law, F. Zhang, N. Wang, Intrinsic valley Hall transport in atomically thin MoS₂, *Nature Communications*, 10 (2019) 611. <https://doi.org/10.1038/s41467-019-08629-9>.
- [8] K. Zollner, P.E.F. Junior, J. Fabian, Strain-tunable orbital, spin-orbit, and optical properties of monolayer transition-metal dichalcogenides, *Physical Review B*, 100 (2019) 195126. <https://doi.org/10.1103/PhysRevB.100.195126>.
- [9] M.M. Islam, V. Chowdhury, M. Nure-Alam-Dipu, A. Zubair, Strain- and Electric-Field-Tunable Valley Polarization in Mo_{0.75}V_{0.25}Te₂ (Mo₃VTe₈) for Valleytronic Application, Jun. 18, 2026, arXiv: arXiv:2606.19954. <https://doi.org/10.48550/arXiv.2606.19954>.
- [10] S. Islam, S.M. Mominuzzaman, A. Zubair, Tunable valley polarization and anomalous Hall effect in ferrovalley NbX₂ and TaX₂ (X = S, Se, Te): A first-principles study, *Applied Surface Science*, 719 (2026) 165094. <https://doi.org/10.1016/j.apsusc.2025.165094>.
- [11] V. Chowdhury, A. Zubair, Strain-tunable spin filtering and valley splitting coexisting with the anomalous Hall effect in the 2D half-metallic VSe₂/VN heterostructure: toward a unified spintronic–valleytronic platform, *Journal of Materials Chemistry C*, 14 (2026) 7794–7809. <https://doi.org/10.1039/D5TC04495B>.
- [12] Y.-L. Hong, Z. Liu, L. Wang, T. Zhou, W. Ma, C. Xu, S. Feng, L. Chen, M.-L. Chen, D.-M. Sun, X.-Q. Chen, H.-M. Cheng, W. Ren, Chemical vapor deposition of layered two-dimensional MoSi₂N₄ materials, *Science*, 369 (2020) 670–674. <https://doi.org/10.1126/science.abb7023>.
- [13] L. Wang, Y. Shi, M. Liu, A. Zhang, Y.-L. Hong, R. Li, Q. Gao, M. Chen, W. Ren, H.-M. Cheng, Y. Li, X.-Q. Chen, Intercalated architecture of MA₂Z₄ family layered van der Waals materials with emerging topological, magnetic and superconducting properties, *Nature Communications*, 12 (2021) 2361. <https://doi.org/10.1038/s41467-021-22324-8>.
- [14] W. Jin, J. Zuo, J. Pang, J. Yang, X. Yu, H. Zhong, X. Kuang, C. Lu, Two-Dimensional MoSi₂N₄ Family: Progress and Perspectives Form Theory, *The Journal of Physical Chemistry Letters*, 15 (2024) 10284–10294. <https://doi.org/10.1021/acs.jpclett.4c02452>.

- [15] T. Latychevskaia, D.A. Bandurin, K.S. Novoselov, A new family of septuple-layer 2D materials of MoSi_2N_4 -like crystals, *Nature Reviews Physics*, 6 (2024) 426–438. <https://doi.org/10.1038/s42254-024-00728-x>.
- [16] S. Li, W. Wu, X. Feng, S. Guan, W. Feng, Y. Yao, S.A. Yang, Valley-dependent properties of monolayer MoSi_2N_4 , WSi_2N_4 , and MoSi_2As_4 , *Physical Review B*, 102 (2020) 235435. <https://doi.org/10.1103/PhysRevB.102.235435>.
- [17] H. Ai, D. Liu, J. Geng, S. Wang, K.H. Lo, H. Pan, Theoretical evidence of the spin–valley coupling and valley polarization in two-dimensional MoSi_2X_4 ($\text{X} = \text{N}, \text{P}$, and As), *Physical Chemistry Chemical Physics*, 23 (2021) 3144–3151. <https://doi.org/10.1039/D0CP05926A>.
- [18] W. Zhou, L. Wu, A. Li, B. Zhang, F. Ouyang, Structural Symmetry, Spin–Orbit Coupling, and Valley-Related Properties of Monolayer WSi_2N_4 Family, *The Journal of Physical Chemistry Letters*, 12 (2021) 11622–11628. <https://doi.org/10.1021/acs.jpclett.1c03197>.
- [19] S.-D. Guo, W.-Q. Mu, Y.-T. Zhu, R.-Y. Han, W.-C. Ren, Predicted septuple-atomic-layer Janus MSiGeN_4 ($\text{M} = \text{Mo}$ and W) monolayers with Rashba spin splitting and high electron carrier mobilities, *Journal of Materials Chemistry C*, 9 (2021) 2464–2473. <https://doi.org/10.1039/D0TC05649A>.
- [20] S. Sheoran, A. Phutela, R. Moulik, S. Bhattacharya, Manipulation of Valley and Spin Properties in Two-Dimensional Janus WSiGeZ_4 ($\text{Z} = \text{N}, \text{P}, \text{As}$) through Symmetry Control, *The Journal of Physical Chemistry C*, 127 (2023) 11396–11406. <https://doi.org/10.1021/acs.jpcc.3c02819>.
- [21] Q. Cui, Y. Zhu, J. Liang, P. Cui, H. Yang, Spin-valley coupling in a two-dimensional VSi_2N_4 monolayer, *Physical Review B*, 103 (2021) 085421. <https://doi.org/10.1103/PhysRevB.103.085421>.
- [22] Y. Liu, T. Zhang, K. Dou, W. Du, R. Peng, Y. Dai, B. Huang, Y. Ma, Valley-Contrasting Physics in Single-Layer CrSi_2N_4 and CrSi_2P_4 , *The Journal of Physical Chemistry Letters*, 12 (2021) 8341–8346. <https://doi.org/10.1021/acs.jpclett.1c02069>.
- [23] B. Mortazavi, F. Shojaei, B. Javvaji, T. Rabczuk, X. Zhuang, Outstandingly high thermal conductivity, elastic modulus, carrier mobility and piezoelectricity in two-dimensional semiconducting CrC_2N_4 : a first-principles study, *Materials Today Energy*, 22 (2021) 100839. <https://doi.org/10.1016/j.mtener.2021.100839>.
- [24] M. Samrat, V. Chowdhury, S. Wang, A. Zubair, Robust Spin Splitting and Strain-Controlled Optical Response in Monolayer CrC_2N_4 for Valleytronic and Optoelectronic Applications, Jun. 15, 2026, arXiv: arXiv:2606.17329. <https://doi.org/10.48550/arXiv.2606.17329>.
- [25] M. Huang, Q. Lang, Y. Wang, X. Guo, Z. Ding, J. Yan, X. Liu, First-principles study on the electronic structure and photocatalytic properties of novel two-dimensional Janus CrXC_2N_4 ($\text{X} = \text{Si}, \text{Ge}$), *Journal of Physics D: Applied Physics*, 57 (2024) 425102. <https://doi.org/10.1088/1361-6463/ad61f8>.
- [26] G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, *Physical Review B*, 54 (1996) 11169–11186. <https://doi.org/10.1103/PhysRevB.54.11169>.
- [27] P.E. Blöchl, Projector augmented-wave method, *Physical Review B*, 50 (1994) 17953–17979. <https://doi.org/10.1103/PhysRevB.50.17953>.
- [28] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized Gradient Approximation Made Simple, *Physical Review Letters*, 77 (1996) 3865–3868. <https://doi.org/10.1103/PhysRevLett.77.3865>.

- [29] J. Heyd, G.E. Scuseria, M. Ernzerhof, Hybrid functionals based on a screened Coulomb potential, *The Journal of Chemical Physics*, 118 (2003) 8207–8215. <https://doi.org/10.1063/1.1564060>.
- [30] A.V. Krukau, O.A. Vydrov, A.F. Izmaylov, G.E. Scuseria, Influence of the exchange screening parameter on the performance of screened hybrid functionals, *The Journal of Chemical Physics*, 125 (2006) 224106. <https://doi.org/10.1063/1.2404663>.
- [31] S. Grimme, Semiempirical GGA-type density functional constructed with a long-range dispersion correction, *Journal of Computational Chemistry*, 27 (2006) 1787–1799. <https://doi.org/10.1002/jcc.20495>.
- [32] G. Henkelman, A. Arnaldsson, H. Jónsson, A fast and robust algorithm for Bader decomposition of charge density, *Computational Materials Science*, 36 (2006) 354–360. <https://doi.org/10.1016/j.commatsci.2005.04.010>.
- [33] A.A. Mostofi, J.R. Yates, Y.-S. Lee, I. Souza, D. Vanderbilt, N. Marzari, wannier90: A tool for obtaining maximally-localised Wannier functions, *Computer Physics Communications*, 178 (2008) 685–699. <https://doi.org/10.1016/j.cpc.2007.11.016>.
- [34] S.S. Tsirkin, High performance Wannier interpolation of Berry curvature and related quantities with WannierBerri code, *npj Computational Materials*, 7 (2021) 33. <https://doi.org/10.1038/s41524-021-00498-5>.
- [35] X. Feng, X. Xu, Z. He, R. Peng, Y. Dai, B. Huang, Y. Ma, Valley-related multiple Hall effect in monolayer VSi₂P, *Physical Review B*, 104 (2021) 075421. <https://doi.org/10.1103/PhysRevB.104.075421>.
- [36] M. Long, F. Miao, M. Xu, S. Feng, Y. Yang, Characterization of ferromagnetic semiconductors and valley polarization in janus VBXS₂ (X = N, P) monolayers, *Physica Scripta*, 99 (2024) 095979. <https://doi.org/10.1088/1402-4896/ad6ec1>.
- [37] B. Zhou, Z. Li, J. Wang, X. Niu, C. Luan, Tunable valley splitting and an anomalous valley Hall effect in hole-doped WS₂ by proximity coupling with a ferromagnetic MnO₂ monolayer, *Nanoscale*, 11 (2019) 13567–13575. <https://doi.org/10.1039/c9nr03315g>.
- [38] D. Xiao, W. Yao, Q. Niu, Valley-Contrasting Physics in Graphene: Magnetic Moment and Topological Transport, *Physical Review Letters*, 99 (2007) 236809. <https://doi.org/10.1103/PhysRevLett.99.236809>.
- [39] N. Nagaosa, J. Sinova, S. Onoda, A.H. MacDonald, N.P. Ong, Anomalous Hall effect, *Reviews of Modern Physics*, 82 (2010) 1539–1592. <https://doi.org/10.1103/RevModPhys.82.1539>.
- [40] E. Liu, Y. Sun, N. Kumar, L. Muechler, A. Sun, L. Jiao, S.-Y. Yang, D. Liu, A. Liang, Q. Xu, J. Kroder, V. Süß, H. Borrmann, C. Shekhar, Z. Wang, C. Xi, W. Wang, W. Schnelle, S. Wirth, Y. Chen, S.T.B. Goennenwein, C. Felser, Giant anomalous Hall effect in a ferromagnetic kagome-lattice semimetal, *Nature Physics*, 14 (2018) 1125–1131. <https://doi.org/10.1038/s41567-018-0234-5>.
- [41] M.W.-Y. Tu, C. Li, H. Yu, W. Yao, Non-adiabatic Hall effect at Berry curvature hot spot, *2D Materials*, 7 (2020) 045004. <https://doi.org/10.1088/2053-1583/ab89e8>.
- [42] G.Y. Guo, S. Murakami, T.-W. Chen, N. Nagaosa, Intrinsic Spin Hall Effect in Platinum: First-Principles Calculations, *Physical Review Letters*, 100 (2008) 096401. <https://doi.org/10.1103/PhysRevLett.100.096401>.
- [43] J. Abir, T. Rahman, M. Khan, S.S. Pallab, R.S. Islam, S.H. Naqib, Phase Switchable Photocatalytic Water Splitting via a Paraelectric-Ferroelectric Transition in Zr₂Ge₂S₆ Monolayer: A Comprehensive Theoretical Insights. 2026. <https://doi.org/10.48550/arXiv.2609.00724>.

- [44] S. Trasatti, The absolute electrode potential: an explanatory note (Recommendations 1986), *Pure and Applied Chemistry*, 58 (1986) 955–966. <https://doi.org/10.1351/pac198658070955>.
- [45] K.-A.N. Duerloo, M.T. Ong, E.J. Reed, Intrinsic Piezoelectricity in Two-Dimensional Materials, *The Journal of Physical Chemistry Letters*, (2012). <https://doi.org/10.1021/jz3012436>.
- [46] M.N. Blonsky, H.L. Zhuang, A.K. Singh, R.G. Hennig, Ab Initio Prediction of Piezoelectricity in Two-Dimensional Materials, *ACS Nano*, 9 (2015) 9885–9891. <https://doi.org/10.1021/acsnano.5b03394>.
- [47] L. Xie, L. Wang, Y. Ma, H. Lu, Y. Yang, Exploring the valleytronic, optical, and piezoelectric properties of Janus MoBXY_2 ($X = \text{N, P}$; $Y = \text{S, Se, Te}$) monolayers for multifunctional applications, *Physical Chemistry Chemical Physics*, 26 (2024) 28474–28483. <https://doi.org/10.1039/d4cp03793f>.
- [48] X. Cai, G. Chen, R. Li, Z. Pan, Y. Jia, Novel valleytronic and piezoelectric properties coexisting in Janus MoAZ_3H ($A = \text{Si, or Ge}$; $Z = \text{N, P, or As}$) monolayers, *Journal of Materials Chemistry C*, 12 (2024) 4682–4689. <https://doi.org/10.1039/D4TC00188E>.
- [49] H.J. Conley, B. Wang, J.I. Ziegler, R.F. Haglund, S.T. Pantelides, K.I. Bolotin, Bandgap Engineering of Strained Monolayer and Bilayer MoS_2 , *Nano Letters*, 13 (2013) 3626–3630. <https://doi.org/10.1021/nl4014748>.
- [50] C.-H. Chang, X. Fan, S.-H. Lin, J.-L. Kuo, Orbital analysis of electronic structure and phonon dispersion in MoS_2 , MoSe_2 , WS_2 , and WSe_2 monolayers under strain, *Physical Review B*, 88 (2013) 195420. <https://doi.org/10.1103/PhysRevB.88.195420>.
- [51] X. Ma, X. Wu, H. Wang, Y. Wang, A Janus MoSSe monolayer: a potential wide solar-spectrum water-splitting photocatalyst with a low carrier recombination rate, *Journal of Materials Chemistry A*, 6 (2018) 2295–2301. <https://doi.org/10.1039/C7TA10015A>.
- [52] R. Peng, Y. Ma, B. Huang, Y. Dai, Two-dimensional Janus PtSSe for photocatalytic water splitting under the visible or infrared light, *Journal of Materials Chemistry A*, 7 (2019) 603–610. <https://doi.org/10.1039/C8TA09177C>.
- [53] P. Wang, Y. Zong, H. Liu, H. Wen, H.-B. Wu, J.-B. Xia, Highly efficient photocatalytic water splitting and enhanced piezoelectric properties of 2D Janus group-III chalcogenides, *Journal of Materials Chemistry C*, 9 (2021) 4989–4999. <https://doi.org/10.1039/D1TC00318F>.
- [54] Y. Fan, B. Yang, X. Song, X. Shao, M. Zhao, Direct Z-scheme photocatalytic overall water splitting on 2D CdS/InSe heterostructures, *Journal of Physics D: Applied Physics*, 51 (2018) 395501. <https://doi.org/10.1088/1361-6463/aad8a2>.
- [55] S.G. Rudi, S. Soleimani-Amiri, A. Rezavand, N. Ghobadi, Enhanced performance of Janus XMSiY_2 ($X = \text{S, Se}$; $M = \text{Mo, W}$; and $Y = \text{N, P}$) monolayers for photocatalytic water splitting via strain engineering, *Journal of Physics and Chemistry of Solids*, 181 (2023) 111561. <https://doi.org/10.1016/j.jpcs.2023.111561>.